

Supporting Information

Future warming enhances rates of population increase of arthropod crop pests globally

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An update version of this document will be submitted with author names and affiliations in case the manuscript is finally accepted for publication.

1. IRTEMP dataset assembly and characteristics:

We searched in Web of Science (<https://www.webofscience.com/>) for published studies from 1900 up to date of searching on 23rd June 2025, following standard guidelines (Siddaway *et al.*, 2019). The searching strategy was as follows:

TS = ((temperat* OR thermal) AND (“life table” OR “life-table” OR “life history” OR “life-history” OR “intrinsic rate of increase” OR demograph* OR “biological parameters” OR “population growth”)) AND

(pest* OR insect* OR aphid* OR caterpillar* OR lepidoptera OR diptera OR hemiptera OR thrips OR larva* OR worm OR nymph OR beetle))

NOT TS = (“natural enem” OR parasitoid OR predator* OR predation OR parasit* OR “biological control” OR forensic OR “medical importance” OR pollinat* OR syrphid* OR hoverfly OR biodegradation OR waste OR seasonal OR abundance)

The first block has three components: the first one for describing the predictor (temperature), a second one with keywords that are typically used in experimental articles aiming to capture our desired response (intrinsic rate of increase), a third part containing refining terms for the biological organisms, combined with the more generic “pest”. The second block discards several terms that would otherwise increase the number of articles returned by the systematic research that lied beyond the scope of this article.

This search yielded 6,136 articles. Then, we applied a first filter by scanning titles of the articles returned to discard those not likely to address variation of population growth rates across lab-controlled temperatures. This first filter allow us to discard 5,158 articles and retain only 978 articles. Next, we applied a second filter by carefully reading the abstracts of these nearly 1,000 standing articles. This enabled us to discard additional 529 articles, leaving only 449.

Finally, we applied a third filter by reading the methods and results sections to only retain articles with more than 4 temperature treatments and their corresponding intrinsic rates of increase r_m values, or alternatively, their doubling time and/or their net fecundity rate R_0 , both of which could be used to infer r_m . After this step that removed another 271 articles, the data set 178 articles. These values of the intrinsic rate of increase follow the Euler-Lotka equations (see e.g., Birch et al. 1953). Data was collected from tables in these papers or, when unavailable in table format, from figures using WebPlotDigitizer v4.8(<https://apps.automeris.io/wpd4/>).

Then, we selected references from previous systematic researches on thermal dependence of r_m that resulted non-redundant with these 178 articles. We used four different data sets from published studies (Gaston, 1988; Frazier *et al.*, 2006 used later by Deutsch *et al.*, 2008; Rezende & Bozinovic, 2019) and systematic research previously carried out by the authors in February 2020 in the Web of Science focused on different performance metrics of arthropod crop pests under warming. After removing duplicates, the resulting data set was definitively named as IRTEMP, including 312 geolocated populations from 241 distinct source articles, 8 taxonomic orders, 56 families and 206 species (**Figure S1**). Details are given in **Table S1**. Of these 312 populations, 30 were shared with Frazier *et al.* (2006) (used later by Deutsch et al. 2008), 3 exclusively with Gaston (1988), 4 with Rezende *et al.* (2019), 51 with our previous systematic research and 224 original from this systematic research.

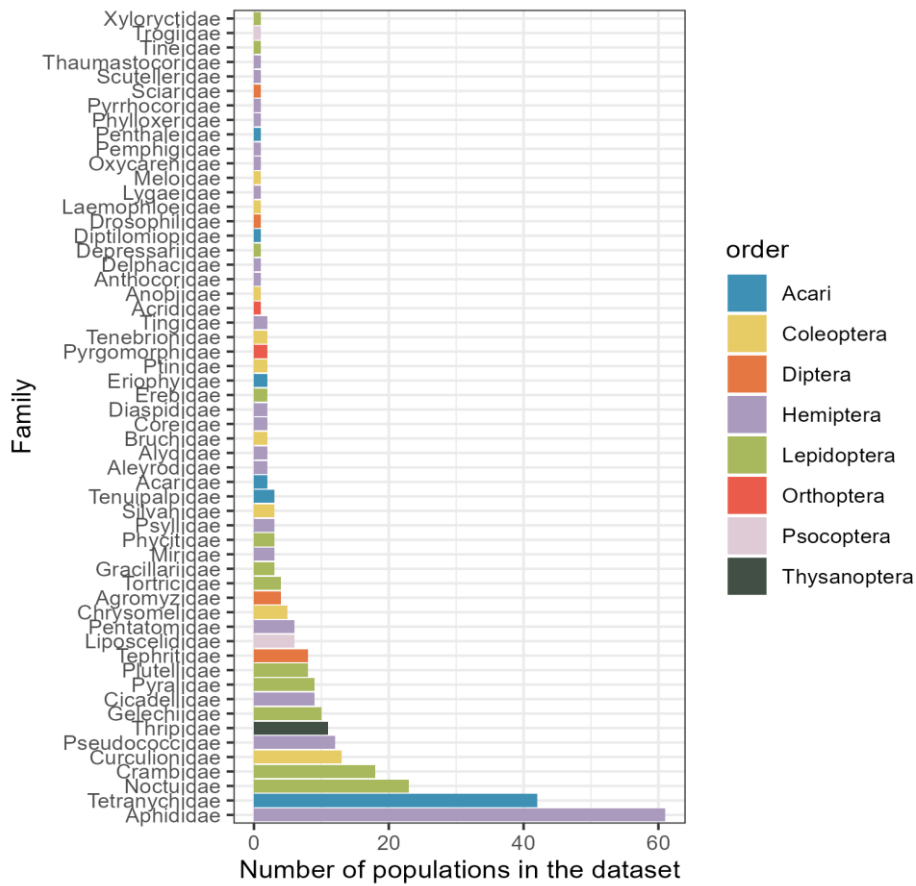


Figure S1: Distribution of the populations in the final data set across taxonomic families and orders

Treatment temperatures

A total of 126 (50.4%) populations were examined at five different temperatures, followed by four ($N = 87$, 27.9%) and 6 ($N = 53$, 17%), and only few were measured at 7 or more temperature treatments (**Figure S2**). These lab-controlled temperatures were almost exclusively at constant regimes ($N = 304$), with few exceptions of either alternating ($N = 1$) or fluctuating ($N = 7$) thermal regimes.

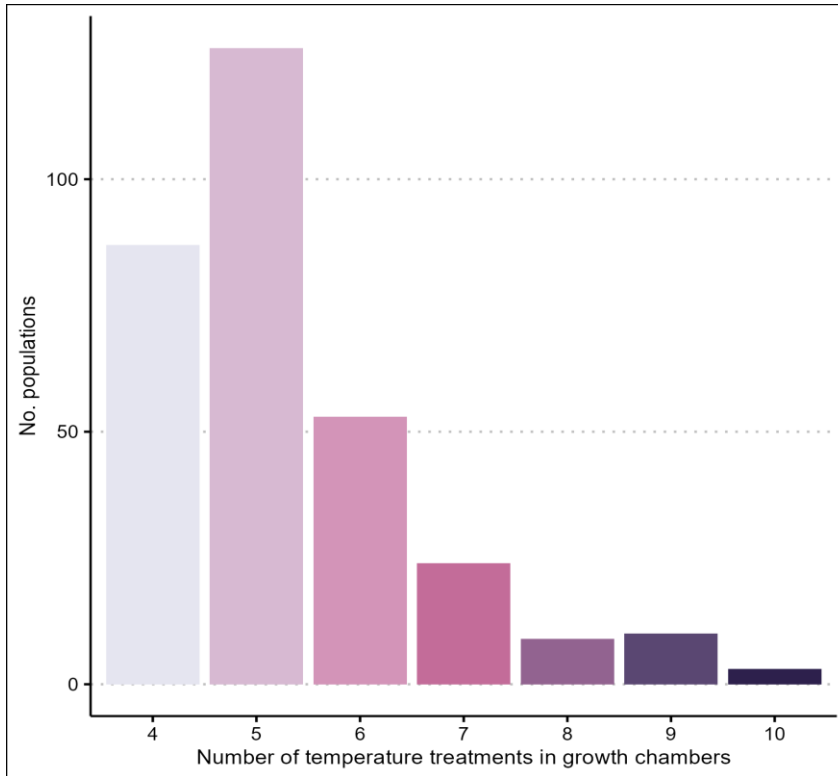


Figure S2: Distribution of the number of controlled temperatures at which populations were measured in source studies

The most repeated temperatures were, in this order: 25, 30, 20, 15 and 35°C (**Figure S3**). Few data were available for temperature extremes (i.e., below 10°C or above 35°C), but the dataset spans temperatures from 5 to 40°C.

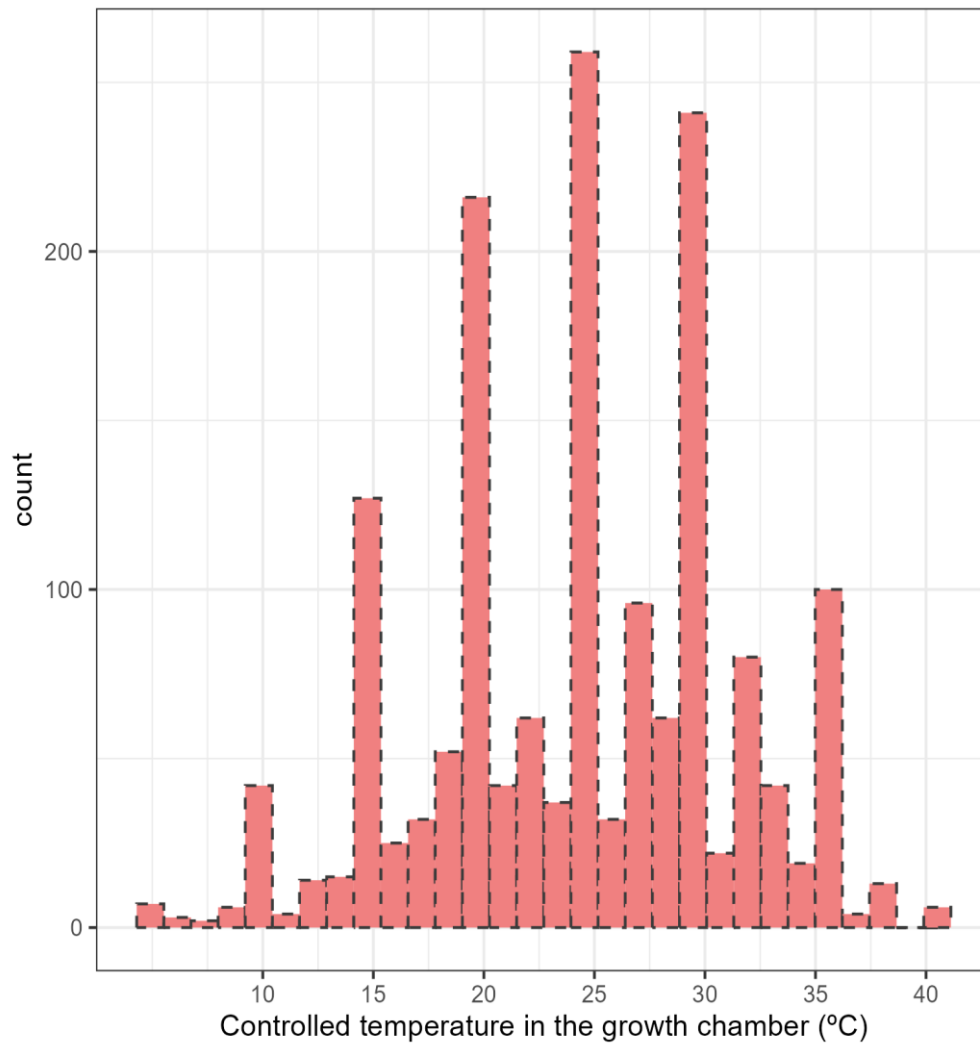


Figure S3: Histogram showing the distribution of the controlled temperatures at which populations were measured in their source studies

Response variable

The intrinsic rates of increase (r_m) had a high variability and a right-skewed distribution (mean = 0.124, median = 0.1; Figure S4).

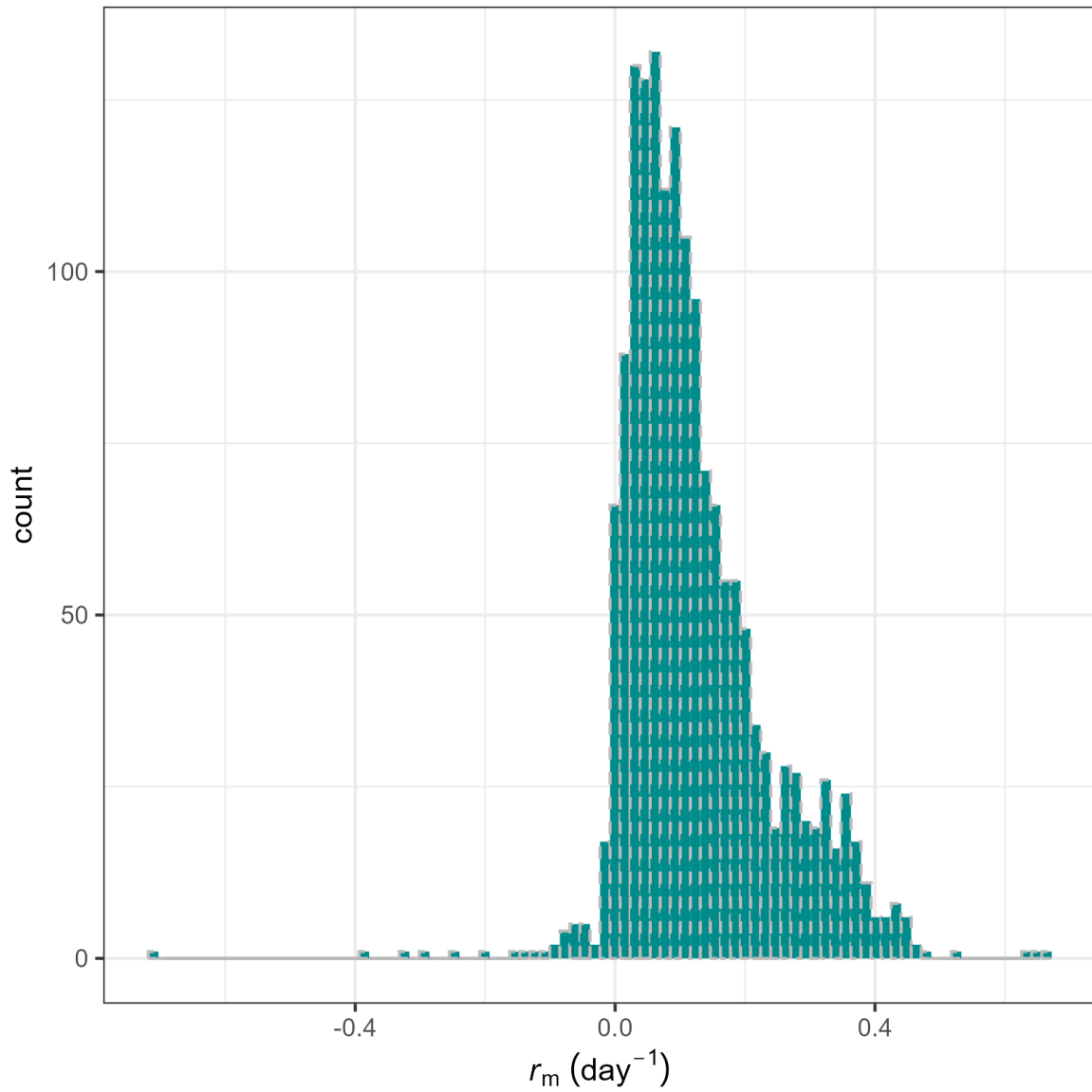


Figure S4: Histogram showing the distribution of the intrinsic rates of increase measured for all the populations in the dataset

Table S1: Taxonomic representation in our dataset and source references for each population (“ID”).

ID	Species	Order : Family	Reference
1	<i>Acanthoscelides obtectus</i>	Coleoptera: Chrysomelidae	(Yao et al., 2016)
2	<i>Acyrtosiphon gossypii</i>	Hemiptera: Aphididae	(Gao et al., 2013)
3	<i>Acyrtosiphon kondoi</i>	Hemiptera: Aphididae	(Summers et al., 1984)
4	<i>Acyrtosiphon kondoi</i>	Hemiptera: Aphididae	(Rohitha & Penman, 1983)
5	<i>Acyrtosiphon kondoi</i>	Hemiptera: Aphididae	(Rohitha & Penman, 1983)
6	<i>Acyrtosiphon pisum</i>	Hemiptera: Aphididae	(Lu & Kuo, 2008)
7	<i>Acyrtosiphon pisum</i>	Hemiptera: Aphididae	(Morgan et al., 2001)
8	<i>Acyrtosiphon pisum</i>	Hemiptera: Aphididae	(Morgan et al., 2001)
9	<i>Acyrtosiphon pisum</i>	Hemiptera: Aphididae	(Campbell & Mackauer, 1977)
10	<i>Adoxophyes orana</i>	Lepidoptera: Tortricidae	(Milonas & Savopoulou-Soultani, 2000)
11	<i>Agriphila aeneociliella</i>	Lepidoptera: Crambidae	(Chi et al., 2016)
12	<i>Aleuroglyphus ovatus</i>	Acari: Acaridae	(Xia et al., 2009)
13	<i>Anaphothrips obscurus</i>	Thysanoptera: Thripidae	(Jiang et al., 2016)
14	<i>Antestiopsis orbitalis</i>	Hemiptera: Pentatomidae	(Azrag et al., 2017)
15	<i>Anthocoris minki pistaciae</i>	Hemiptera: Anthocoridae	(Pourali et al., 2012)
16	<i>Anthonomus eugenii</i>	Coleoptera: Curculionidae	(Toapanta et al., 2005)
17	<i>Antigastra catalaunalis</i>	Lepidoptera: Crambidae	(Al-Gerrawy & Al-Shammary, 2020)
18	<i>Aphis citricola</i>	Hemiptera: Aphididae	(Komazaki, 1982)
19	<i>Aphis fabae</i>	Hemiptera: Aphididae	(Akca et al., 2015)
20	<i>Aphis glycines</i>	Hemiptera: Aphididae	(McCornack et al., 2004)
21	<i>Aphis glycines</i>	Hemiptera: Aphididae	(Tian et al., 2020)
22	<i>Aphis gossypii</i>	Hemiptera: Aphididae	(Kocourek et al., 1994)
23	<i>Aphis gossypii</i>	Hemiptera: Aphididae	(Komazaki, 1982)
24	<i>Aphis gossypii</i>	Hemiptera: Aphididae	(Xia et al., 1999)
25	<i>Aphis gossypii</i>	Hemiptera: Aphididae	(Parajulee, 2007)
26	<i>Aphis gossypii</i>	Hemiptera: Aphididae	(Zamani et al., 2006)
27	<i>Aphis nasturtii</i>	Hemiptera: Aphididae	(Wang et al., 1997)
28	<i>Apomyelois ceratoniae</i>	Lepidoptera: Pyralidae	(Eyidozehi et al., 2023)
29	<i>Arimania komaroffi</i>	Lepidoptera: Pyralidae	(Basirat et al., 2015)
30	<i>Athetis dissimilis</i>	Lepidoptera: Noctuidae	(Guo et al., 2017)
31	<i>Atractomorpha lata</i>	Orthoptera: Pyrgomorphidae	(Li et al., 2024)
32	<i>Atractomorpha sinensis</i>	Orthoptera: Pyrgomorphidae	(Li et al., 2020)
33	<i>Aulacorthum solani</i>	Hemiptera: Aphididae	(Jandricic et al., 2010)
34	<i>Aulacorthum solani</i>	Hemiptera: Aphididae	(Lee, Sang guei et al., 2008)
35	<i>Bactrocera cucurbitae</i>	Diptera: Tephritidae	(Vargas et al., 1997)
36	<i>Bactrocera cucurbitae</i>	Diptera: Tephritidae	(Yang et al., 1994)
37	<i>Bactrocera dorsalis</i>	Diptera: Tephritidae	(Vargas et al., 1997)
38	<i>Bactrocera dorsalis</i>	Diptera: Tephritidae	(Yang et al., 1994)
39	<i>Bactrocera latifrons</i>	Diptera: Tephritidae	(Vargas et al., 1997)

ID	Species	Order : Family	Reference
40	<i>Bactrocera scutellata</i>	Diptera: Tephritidae	(Jeon et al., 2012)
41	<i>Bactrocera zonata</i>	Diptera: Tephritidae	(Choudhary et al., 2020)
42	<i>Bemisia argentifolii</i>	Hemiptera: Aleyrodidae	(Wang & Tsai, 1996)
43	<i>Bemisia tabaci</i>	Hemiptera: Aleyrodidae	(Bonato et al., 2007)
44	<i>Brachmia macrocospa</i>	Lepidoptera: Gelechiidae	(Ma et al., 2017)
45	<i>Brachycaudus schwartzi</i>	Hemiptera: Aphididae	(Satar & Yokomi, 2002)
46	<i>Brachynema germari</i>	Hemiptera: Pentatomidae	(Yazdanpanah et al., 2019)
47	<i>Bradysia odoriphaga</i>	Diptera: Sciaridae	(Li et al., 2015)
48	<i>Brevicoryne brassicae</i>	Hemiptera: Aphididae	(Soh et al., 2018)
49	<i>Brevicoryne brassicae</i>	Hemiptera: Aphididae	(Kim et al., 2014)
50	<i>Brevicoryne brassicae</i>	Hemiptera: Aphididae	(SATAR et al., 2005)
51	<i>Brevicoryne brassicae</i>	Hemiptera: Aphididae	(Deloach, 1974)
52	<i>Bryobia praetiosa</i>	Acari: Tetranychidae	(Ullah et al., 2020)
53	<i>Busseola fusca</i>	Lepidoptera: Noctuidae	(Khadioli et al., 2014b)
54	<i>Cactoblastis cactorum</i>	Lepidoptera: Pyralidae	(Legaspi & Legaspi, 2007)
55	<i>Cadra cautella</i>	Lepidoptera: Pyralidae	(Lin et al., 2023)
56	<i>Calliptamus italicus</i>	Orthoptera: Acrididae	(Ren et al., 2016)
57	<i>Callosobruchus machulatus</i>	Coleoptera: Chrysomelidae	(Giga & Smith, 1983)
58	<i>Callosobruchus machulatus</i>	Coleoptera: Chrysomelidae	(Daglish et al., 2021)
59	<i>Callosobruchus machulatus</i>	Coleoptera: Chrysomelidae	(Daglish et al., 2021)
60	<i>Callosobruchus rhodesianus</i>	Coleoptera: Chrysomelidae	(Giga & Smith, 1983)
61	<i>Cassida vittata</i>	Coleoptera: Chrysomelidae	(El Aalaoui & Sbaghi, 2024b)
62	<i>Celepitrimerus vitis</i>	Acari: Eriophyidae	(Walton et al., 2010)
63	<i>Cenopalpus irani</i>	Acari: Tenuipalpidae	(Bazgir et al., 2015)
64	<i>Ceratitis capitata</i>	Diptera: Tephritidae	(Vargas et al., 1997)
65	<i>Chilo orichalcociliellus</i>	Lepidoptera: Crambidae	(Mbapila et al., 2002)
66	<i>Chilo orichalcociliellus</i>	Lepidoptera: Crambidae	(Mbapila et al., 2002)
67	<i>Chilo partellus</i>	Lepidoptera: Crambidae	(Khadioli et al., 2014a)
68	<i>Chilo partellus</i>	Lepidoptera: Crambidae	(Singh, 1991)
69	<i>Chilo partellus</i>	Lepidoptera: Crambidae	(Mbapila et al., 2002)
70	<i>Chilo partellus</i>	Lepidoptera: Crambidae	(Mbapila et al., 2002)
71	<i>Chrysodeixis chalcites</i>	Lepidoptera: Noctuidae	(Pino et al., 2020)
72	<i>Chrysodeixis includens</i>	Lepidoptera: Noctuidae	(Pasqualotto et al., 2024)
73	<i>Cicadulina bipunctata</i>	Hemiptera: Cicadellidae	(Tokuda & Matsumura, 2005)
74	<i>Cinara tujafilina</i>	Hemiptera: Aphididae	(Durak & Borowiak-Sobkowiak, 2013)
75	<i>Clavigralla shadabi</i>	Hemiptera: Coreidae	(Dreyer & Baumgärtner, 1996)
76	<i>Clavigralla tomentosicollis</i>	Hemiptera: Coreidae	(Dreyer & Baumgärtner, 1996)
77	<i>Cnaphalocrocis medinalis</i>	Lepidoptera: Crambidae	(Chang & Wu, 1988)
78	<i>Cnaphalocrocis medinalis</i>	Lepidoptera: Crambidae	(Chang & Wu, 1988)
79	<i>Cnaphalocrocis medinalis</i>	Lepidoptera: Crambidae	(Chang & Wu, 1988)
80	<i>Conogethes punctiferalis</i>	Lepidoptera: Crambidae	(Pang et al., 2022)

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81	<i>Copitarsia decolora</i>	Lepidoptera: Noctuidae	(Gould <i>et al.</i> , 2005)
82	<i>Copitarsia decolora</i>	Lepidoptera: Noctuidae	(Gould <i>et al.</i> , 2005)
83	<i>Cryptolestes ferrugineus</i>	Coleoptera: Laemophloeidae	(Smith, 1965)
84	<i>Cydia pomonella</i>	Lepidoptera: Tortricidae	(Aghdam <i>et al.</i> , 2009)
85	<i>Daktulosphaira vitifoliae</i>	Hemiptera: Phylloxeridae	(Granett & Timper, 1987)
86	<i>Dalbulus elimatus</i>	Hemiptera: Cicadellidae	(Madden <i>et al.</i> , 1986)
87	<i>Dalbulus elimatus</i>	Hemiptera: Cicadellidae	(Madden <i>et al.</i> , 1986)
88	<i>Dalbulus gelbus</i>	Hemiptera: Cicadellidae	(Madden <i>et al.</i> , 1986)
89	<i>Dalbulus maidis</i>	Hemiptera: Cicadellidae	(Madden <i>et al.</i> , 1986)
90	<i>Diaphania indica</i>	Lepidoptera: Crambidae	(Hosseininejad & Naseri, 2015)
91	<i>Diaphorina citri</i>	Hemiptera: Psyllidae	(Liu & Tsai, 2000)
92	<i>Diaspis echinocacti</i>	Hemiptera: Diaspididae	(El Aalaoui & Sbaghi, 2024a)
93	<i>Diatraea lineolata</i>	Lepidoptera: Pyralidae	(Rodríguez-Del-Bosque <i>et al.</i> , 1989)
94	<i>Drosophila suzukii</i>	Diptera: Drosophilidae	(Schlesener <i>et al.</i> , 2020)
95	<i>Dysdercus fasciatus</i>	Hemiptera: Pyrrhocoridae	(Clarke & Sardesai, 1959)
96	<i>Dysmicoccus neobrevipes</i>	Hemiptera: Pseudococcidae	(Qin <i>et al.</i> , 2013)
97	<i>Elasmopalpus lignosellus</i>	Lepidoptera: Pyralidae	(Sandhu <i>et al.</i> , 2013)
98	<i>Eotetranychus carpini</i>	Acari: Tetranychidae	(Bounfour & Tanigoshi, 2001)
99	<i>Eotetranychus carpini</i>	Acari: Tetranychidae	(Bonato <i>et al.</i> , 1990)
100	<i>Eotetranychus orientalis</i>	Acari: Tetranychidae	(Wang <i>et al.</i> , 2024)
101	<i>Eotetranychus sexmaculatus</i>	Acari: Tetranychidae	(Wang <i>et al.</i> , 2024)
102	<i>Eotetranychus frosti</i>	Acari: Tetranychidae	(Jafarian & Jafari, 2016)
103	<i>Ephestia kuehniella</i>	Lepidoptera: Phycitidae	(Siddiqui & Barlow, 1973)
104	<i>Ephestia kuehniella</i>	Lepidoptera: Phycitidae	(Imura, 1986)
105	<i>Ephestia kuehniella</i>	Lepidoptera: Phycitidae	(Imura, 1986)
106	<i>Ephestia kuehniella</i>	Lepidoptera: Phycitidae	(Imura, 1986)
107	<i>Epicauta impressicornis</i>	Coleoptera: Meloidae	(Liu <i>et al.</i> , 2018)
108	<i>Epiphyas postvittana</i>	Lepidoptera: Tortricidae	(Danthanarayana <i>et al.</i> , 1995)
109	<i>Eriosoma lanigerum</i>	Hemiptera: Aphididae	(Asante <i>et al.</i> , 1991)
110	<i>Euphyllura olivina</i>	Hemiptera: Psyllidae	(El Aalaoui <i>et al.</i> , 2025)
111	<i>Eurygaster integriceps</i>	Hemiptera: Scutelleridae	(Iranipour <i>et al.</i> , 2010)
112	<i>Euschistus heros</i>	Hemiptera: Pentatomidae	(Rodrigues <i>et al.</i> , 2023)
113	<i>Eutetranychus africanus</i>	Acari: Tetranychidae	(Lin <i>et al.</i> , 2020)
114	<i>Eutetranychus banksi</i>	Acari: Tetranychidae	(Childers <i>et al.</i> , 1991)
115	<i>Frankliniella busca</i>	Thysanoptera: Thripidae	(Lowry <i>et al.</i> , 1992)
116	<i>Galerucella irmanica</i>	Coleoptera: Chrysomelidae	(Zheng <i>et al.</i> , 2008)
117	<i>Grapholita molesta</i>	Lepidoptera: Tortricidae	(Chen <i>et al.</i> , 2019)
118	<i>Halyomorpha halys</i>	Hemiptera: Pentatomidae	(Mermer <i>et al.</i> , 2023)
119	<i>Halyomorpha halys</i>	Hemiptera: Pentatomidae	(Mermer <i>et al.</i> , 2023)
120	<i>Halyomorpha halys</i>	Hemiptera: Pentatomidae	(Govindan & Hutchison, 2020)
121	<i>Helicoverpa armigera</i>	Lepidoptera: Noctuidae	(Dalal & Arora, 2021)

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122	<i>Helicoverpa armigera</i>	Lepidoptera: Noctuidae	(Noor-ul-Ane et al., 2018)
123	<i>Helicoverpa armigera</i>	Lepidoptera: Noctuidae	(Mironidis, 2014)
124	<i>Helicoverpa armigera</i>	Lepidoptera: Noctuidae	(Mironidis & Savopoulou-Soultani, 2008)
125	<i>Helicoverpa armigera</i>	Lepidoptera: Noctuidae	(Mironidis & Savopoulou-Soultani, 2008)
126	<i>Hyadaphis foeniculi</i>	Hemiptera: Aphididae	(Ramalho et al., 2015)
127	<i>Hyadaphis pseudobrassicae</i>	Hemiptera: Aphididae	(Deloach, 1974)
128	<i>Hyalopterus pruni</i>	Hemiptera: Aphididae	(Latham & Mills, 2011)
129	<i>Hypera postica</i>	Coleoptera: Curculionidae	(Zahiri et al., 2010)
130	<i>Hypothenemus hampei</i>	Coleoptera: Curculionidae	(Wei et al., 2023)
131	<i>Hypothenemus hampei</i>	Coleoptera: Curculionidae	(Jaramillo et al., 2009)
132	<i>Jakowleffia setulosa</i>	Hemiptera: Oxycarenidae	(He et al., 2014)
133	<i>Lasioderma serricorne</i>	Coleoptera: Ptinidae	(Wang et al., 2021)
134	<i>Lasioderma serricorne</i>	Coleoptera: Ptinidae	(Howe, 1957)
135	<i>Lepinotus reticulatus</i>	Psocoptera: Anobiidae	(Opit et al., 2010)
136	<i>Leucinodes orbonalis</i>	Lepidoptera: Crambidae	(Islam et al., 2020)
137	<i>Liposcelis badia</i>	Psocoptera: Liposcelididae	(Jiang et al., 2008)
138	<i>Liposcelis bostrychophila</i>	Psocoptera: Liposcelididae	(Wang et al., 2000)
139	<i>Liposcelis decolor</i>	Psocoptera: Liposcelididae	(Tang et al., 2008)
140	<i>Liposcelis paeta</i>	Psocoptera: Liposcelididae	(Wang et al., 2009)
141	<i>Liposcelis tricolor</i>	Psocoptera: Liposcelididae	(Dong et al., 2007)
142	<i>Liposcelis yunnanensis</i>	Psocoptera: Liposcelididae	(Hassan et al., 2011)
143	<i>Liriomyza sativae</i>	Diptera: Agromyzidae	(Wang, 2014)
144	<i>Liriomyza sativae</i>	Diptera: Agromyzidae	(He et al., 1999)
145	<i>Liriomyza sativae</i>	Diptera: Agromyzidae	(Haghani et al., 2007)
146	<i>Liriomyza trifolii</i>	Diptera: Agromyzidae	(Wang, 2014)
147	<i>Lygus hesperus</i>	Hemiptera: Miridae	(Strong & Sheldahl, 1970)
148	<i>Lygus lineolaris</i>	Hemiptera: Miridae	(Ugine, 2012)
149	<i>Lygus rugulipennis</i>	Hemiptera: Miridae	(Zamanzadeh et al., 2022)
150	<i>Lymantria nebulosa</i>	Lepidoptera: Erebidae	(Lei et al., 2021)
151	<i>Macrosiphum euphorbiae</i>	Hemiptera: Aphididae	(Zhou et al., 2014)
152	<i>Macrosiphum euphorbiae</i>	Hemiptera: Aphididae	(Barlow, 1962)
153	<i>Macrosiphum rosae</i>	Hemiptera: Aphididae	(Olmez et al., 2003)
154	<i>Marasmia exigua</i>	Lepidoptera: Pyralidae	(Liao et al., 2017)
155	<i>Matsumuratettix hiroglyphicus</i>	Hemiptera: Cicadellidae	(Kobori & Hanboonsong, 2017)
156	<i>Melanaphis sacchari</i>	Hemiptera: Aphididae	(Souza & Davis, 2020)
157	<i>Monosteira unicostata</i>	Hemiptera: Tingidae	(Sánchez-Ramos et al., 2017)
158	<i>Mythimna loreyi</i>	Lepidoptera: Noctuidae	(Qin et al., 2017)
159	<i>Myzus persicae</i>	Hemiptera: Aphididae	(Ahn et al., 2025)
160	<i>Myzus persicae</i>	Hemiptera: Aphididae	(Ahn et al., 2025)
161	<i>Myzus persicae</i>	Hemiptera: Aphididae	(Li et al., 2025)
162	<i>Myzus persicae</i>	Hemiptera: Aphididae	(Zhou et al., 2014)

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163	<i>Myzus persicae</i>	Hemiptera: Aphididae	(Barbosa <i>et al.</i> , 2011)
164	<i>Myzus persicae</i>	Hemiptera: Aphididae	(Davis <i>et al.</i> , 2006)
165	<i>Myzus persicae</i>	Hemiptera: Aphididae	(Davis <i>et al.</i> , 2006)
166	<i>Myzus persicae</i>	Hemiptera: Aphididae	(Din, 1976)
167	<i>Myzus persicae</i>	Hemiptera: Aphididae	(Barlow, 1962)
168	<i>Myzus persicae</i>	Hemiptera: Aphididae	(Deloach, 1974)
169	<i>Nasonovia ribisnigri</i>	Hemiptera: Aphididae	(Diaz & Fereres, 2005)
170	<i>Nephotettix cincticeps</i>	Hemiptera: Cicadellidae	(Valle <i>et al.</i> , 1986)
171	<i>Nephotettix malayanus</i>	Hemiptera: Cicadellidae	(Valle <i>et al.</i> , 1986)
172	<i>Nephotettix nigropictus</i>	Hemiptera: Cicadellidae	(Valle <i>et al.</i> , 1986)
173	<i>Nephotettix nigropictus</i>	Hemiptera: Cicadellidae	(Valle <i>et al.</i> , 1986)
174	<i>Nilaparvarta lugens</i>	Hemiptera: Delphacidae	(Sahoo <i>et al.</i> , 2025)
175	<i>Oligonychus biharensis</i>	Acari: Tetranychidae	(Wang <i>et al.</i> , 2024)
176	<i>Oligonychus litchii</i>	Acari: Tetranychidae	(Chen <i>et al.</i> , 2016)
177	<i>Oligonychus litchii</i>	Acari: Tetranychidae	(Quan <i>et al.</i> , 2019)
178	<i>Oligonychus mangiferus</i>	Acari: Tetranychidae	(Lin, 2013)
179	<i>Opisina arenosella</i>	Lepidoptera: Xyloryctidae	(Lu <i>et al.</i> , 2016)
180	<i>Opogona sacchari</i>	Lepidoptera: Tineidae	(Fonseca Lacerda <i>et al.</i> , 2019)
181	<i>Oryzaephilus surinamensis</i>	Coleoptera: Silvanidae	(Beckett & Evans, 1994)
182	<i>Oryzaephilus surinamensis</i>	Coleoptera: Silvanidae	(Beckett & Evans, 1994)
183	<i>Oryzaephilus surinamensis</i>	Coleoptera: Silvanidae	(Beckett & Evans, 1994)
184	<i>Palpita forficifera</i>	Lepidoptera: Crambidae	(Scheunemann <i>et al.</i> , 2021)
185	<i>Palpita unionalis</i>	Lepidoptera: Crambidae	(Mansour, 2024)
186	<i>Panonychus ulmi</i>	Acari: Tetranychidae	(Han <i>et al.</i> , 2007)
187	<i>Paraponyx crisonalis</i>	Lepidoptera: Crambidae	(Chen <i>et al.</i> , 2017)
188	<i>Paromius exiguus</i>	Hemiptera: Lygaeidae	(Park <i>et al.</i> , 2017)
189	<i>Pectinophora gossypiella</i>	Lepidoptera: Gelechiidae	(Nagaraju <i>et al.</i> , 2024)
190	<i>Penthaleus major</i>	Acari: Penthaleidae	(Wang <i>et al.</i> , 2023)
191	<i>Pezothrips kellyanus</i>	Thysanoptera: Thripidae	(Varikou <i>et al.</i> , 2012)
192	<i>Phenacoccus manihoti</i>	Hemiptera: Pseudococcidae	(Lema & Herren, 1985)
193	<i>Phenacoccus solenopsis</i>	Hemiptera: Pseudococcidae	(Harbi <i>et al.</i> , 2025)
194	<i>Phenacoccus solenopsis</i>	Hemiptera: Pseudococcidae	(Fand <i>et al.</i> , 2014)
195	<i>Phenacoccus solenopsis</i>	Hemiptera: Pseudococcidae	(Chen <i>et al.</i> , 2015)
196	<i>Phenacoccus solenopsis</i>	Hemiptera: Pseudococcidae	(Chen <i>et al.</i> , 2015)
197	<i>Phenacoccus solenopsis</i>	Hemiptera: Pseudococcidae	(Kumar <i>et al.</i> , 2013)
198	<i>Phenacoccus solenopsis</i>	Hemiptera: Pseudococcidae	(Chen <i>et al.</i> , 2015)
199	<i>Phenacoccus solenopsis</i>	Hemiptera: Pseudococcidae	(Kumar <i>et al.</i> , 2013)
200	<i>Phthorimaea operculella</i>	Lepidoptera: Gelechiidae	(Golizadeh <i>et al.</i> , 2012)
201	<i>Phyllocnistis citrella</i>	Lepidoptera: Gracillariidae	(Elekçioğlu & Uygün, 2004)
202	<i>Phyllocoptruta oleivora</i>	Acari: Eriophyidae	(Allen <i>et al.</i> , 1995)
203	<i>Phyllonorycter corylifoliella</i>	Lepidoptera: Gracillariidae	(Amiri <i>et al.</i> , 2010)

ID	Species	Order : Family	Reference
204	<i>Phyllonorycter ringoniella</i>	Lepidoptera: Gracillariidae	(Geng & Jung, 2017)
205	<i>Planococcus citri</i>	Hemiptera: Pseudococcidae	(Satar & Karacaoğlu, 2017)
206	<i>Plutella xylostella</i>	Lepidoptera: Plutellidae	(Saeed et al., 2019)
207	<i>Plutella xylostella</i>	Lepidoptera: Plutellidae	(Dan et al., 1995; Shi et al., 2012)
208	<i>Plutella xylostella</i>	Lepidoptera: Plutellidae	(Marchioro & Foerster, 2012)
209	<i>Plutella xylostella</i>	Lepidoptera: Plutellidae	(Ngowi et al., 2017)
210	<i>Plutella xylostella</i>	Lepidoptera: Plutellidae	(Golizadeh et al., 2009)
211	<i>Plutella xylostella</i>	Lepidoptera: Plutellidae	(Golizadeh et al., 2009)
212	<i>Plutella xylostella</i>	Lepidoptera: Plutellidae	(Sarnthoy et al., 1989)
213	<i>Plutella xylostella</i>	Lepidoptera: Plutellidae	(Sarnthoy et al., 1989)
214	<i>Pseudaletia separata</i>	Lepidoptera: Plutellidae	(Ko et al., 2003)
215	<i>Pseudaulacaspis pentagona</i>	Hemiptera: Diaspididae	(Erkiliç & Uygun, 1997)
216	<i>Pseudococcus comstocki</i>	Hemiptera: Pseudococcidae	(Heidari, 1999)
217	<i>Psylla chinensis</i>	Hemiptera: Psyllidae	(Zhang et al., 2024)
218	<i>Pterohelaeus alternatus</i>	Coleoptera: Tenebrionidae	(Allsopp, 1981)
219	<i>Pterohelaeus darlingiensis</i>	Coleoptera: Tenebrionidae	(Allsopp, 1981)
220	<i>Ptinus tectus</i>	Coleoptera: Ptinidae	(Howe, 1953)
221	<i>Rachiplusia un</i>	Lepidoptera: Noctuidae	(Pasqualotto et al., 2024)
222	<i>Raoiella indica</i>	Acari: Tenuipalpidae	(Fidelis et al., 2019)
223	<i>Raoiella indica</i>	Acari: Tenuipalpidae	(Mejía et al., 2019)
224	<i>Rhizopertha dominica</i>	Coleoptera: Curculionidae	(Birch, 1953)
225	<i>Rhizopertha dominica</i>	Coleoptera: Curculionidae	(Birch, 1953)
226	<i>Rhizopertha dominica</i>	Coleoptera: Curculionidae	(Birch, 1953)
227	<i>Rhopalosiphum padi</i>	Hemiptera: Aphididae	(Asin & Pons, 2001)
228	<i>Rhopalosiphum padi</i>	Hemiptera: Aphididae	(Aquad et al., 2009)
229	<i>Rhopalosiphum rufiabdominalis</i>	Hemiptera: Aphididae	(Tsai & Liu, 1998)
230	<i>Rhyncaphytoptus ficifoliae</i>	Acari: Dptilomiopidae	(Bahrae et al., 2019)
231	<i>Rynchophorus ferrugineus</i>	Coleoptera: Curculionidae	(Peng et al., 2016)
232	<i>Riptortus pedestris</i>	Hemiptera: Alydidae	(Tian et al., 2022)
233	<i>Riptortus pedestris</i>	Hemiptera: Alydidae	(Ahn et al., 2019)
234	<i>Saccharicoccus sacchari</i>	Hemiptera: Pseudococcidae	(Rae & Death, 1991)
235	<i>Schizaphis graminum</i>	Hemiptera: Aphididae	(Pendleton et al., 2009)
236	<i>Schizaphis graminum</i>	Hemiptera: Aphididae	(Pendleton et al., 2009)
237	<i>Scirtothrips perseae</i>	Thysanoptera: Thripidae	(Hoddle, 2002)
238	<i>Sesamia calamistis</i>	Lepidoptera: Noctuidae	(Khadioli et al., 2014b)
239	<i>Sesamia cretica</i>	Lepidoptera: Noctuidae	(Arbabafti et al., 2021)
240	<i>Sesamia cretica</i>	Lepidoptera: Noctuidae	(Arbabafti et al., 2021)
241	<i>Sipha mayidis</i>	Hemiptera: Aphididae	(Taylor et al., 2023)
242	<i>Sitobion avenae</i>	Hemiptera: Aphididae	(Asin & Pons, 2001)
243	<i>Sitobion fragariae</i>	Hemiptera: Aphididae	(Turak et al., 1998)
244	<i>Sitobion miscanthi</i>	Hemiptera: Aphididae	(Turak et al., 1998)

ID	Species	Order : Family	Reference
245	<i>Sitophilus oryzae</i>	Coleoptera: Curculionidae	(Lu et al., 2024)
246	<i>Sitophilus oryzae</i>	Coleoptera: Curculionidae	(Birch, 1953)
247	<i>Sitophilus oryzae</i>	Coleoptera: Curculionidae	(Birch, 1953)
248	<i>Sitophilus oryzae</i>	Coleoptera: Curculionidae	(Birch, 1953)
249	<i>Sitophilus oryzae</i>	Coleoptera: Curculionidae	(Birch, 1953)
250	<i>Sitotroga cerealella</i>	Lepidoptera: Gelechiidae	(Hansen et al., 2004)
251	<i>Sitotroga cerealella</i>	Lepidoptera: Gelechiidae	(Hansen et al., 2004)
252	<i>Spilosoma obliqua</i>	Lepidoptera: Erebiidae	(Mobarak et al., 2022)
253	<i>Spodoptera frugiperda</i>	Lepidoptera: Noctuidae	(Tata et al., 2025)
254	<i>Spodoptera frugiperda</i>	Lepidoptera: Noctuidae	(Ouda, et al., 2022)
255	<i>Spodoptera frugiperda</i>	Lepidoptera: Noctuidae	(Prasad et al., 2022)
256	<i>Spodoptera litura</i>	Lepidoptera: Noctuidae	(Rao & Prasad, 2020)
257	<i>Spodoptera litura</i>	Lepidoptera: Noctuidae	(Srinivasa Rao et al., 2014)
258	<i>Spodoptera litura</i>	Lepidoptera: Noctuidae	(Manimanjari et al., 2014)
259	<i>Stenoma impressella</i>	Lepidoptera: Depressariidae	(Martínez et al., 2014)
260	<i>Stephanitis pyri</i>	Hemiptera: Tingidae	(Aysal & Kivan, 2008)
261	<i>Tetraneura nigriabdominalis</i>	Hemiptera: Pemphigidae	(Kuo et al., 2006)
262	<i>Tetranychus cinnabarinus</i>	Acari: Tetranychidae	(He et al., 2008)
263	<i>Tetranychus cinnabarinus</i>	Acari: Tetranychidae	(He et al., 2008)
264	<i>Tetranychus cinnabarinus</i>	Acari: Tetranychidae	(He et al., 2008)
265	<i>Tetranychus evansi</i>	Acari: Tetranychidae	(Bonato, 1999)
266	<i>Tetranychus evansi</i>	Acari: Tetranychidae	(De Moraes & McMurtry, 1987)
267	<i>Tetranychus evansi</i>	Acari: Tetranychidae	(Gotoh et al., 2010)
268	<i>Tetranychus evansi</i>	Acari: Tetranychidae	(Gotoh et al., 2010)
269	<i>Tetranychus evansi</i>	Acari: Tetranychidae	(Gotoh et al., 2010)
270	<i>Tetranychus evansi</i>	Acari: Tetranychidae	(Gotoh et al., 2010)
271	<i>Tetranychus evansi</i>	Acari: Tetranychidae	(Gotoh et al., 2010)
272	<i>Tetranychus evansi</i>	Acari: Tetranychidae	(Gotoh et al., 2010)
273	<i>Tetranychus evansi</i>	Acari: Tetranychidae	(Gotoh et al., 2010)
274	<i>Tetranychus fijiensis</i>	Acari: Tetranychidae	(Veerlapati & Srinivasa, 2023)
275	<i>Tetranychus kanzawai</i>	Acari: Tetranychidae	(Ullah et al., 2011)
276	<i>Tetranychus kanzawai</i>	Acari: Tetranychidae	(Hasanvand et al., 2020)
277	<i>Tetranychus ludeni</i>	Acari: Tetranychidae	(Ristyadi et al., 2021)
278	<i>Tetranychus ludeni</i>	Acari: Tetranychidae	(Gotoh et al., 2015; Nachman & Gotoh, 2015)
279	<i>Tetranychus mcdanieli</i>	Acari: Tetranychidae	(TANIGOSHI et al., 1975; Logan et al., 1976)
280	<i>Tetranychus mcdanieli</i>	Acari: Tetranychidae	(Roy et al., 2003)
281	<i>Tetranychus merganser</i>	Acari: Tetranychidae	(Ullah et al., 2011)
282	<i>Tetranychus piercei</i>	Acari: Tetranychidae	(Gotoh et al., 2015; Nachman & Gotoh, 2015)
283	<i>Tetranychus pueraricola</i>	Acari: Tetranychidae	(Gotoh et al., 2004)

ID	Species	Order : Family	Reference
284	<i>Tetranychus truncatus</i>	Acari: Tetranychidae	(Gotoh <i>et al.</i> , 2015; Nachman & Gotoh, 2015)
285	<i>Tetranychus turkestanii</i>	Acari: Tetranychidae	(Karami-Jamouri & Shishehbor, 2012)
286	<i>Tetranychus urticae</i>	Acari: Tetranychidae	(Gotoh <i>et al.</i> , 2015; Nachman & Gotoh, 2015)
287	<i>Tetranychus urticae</i>	Acari: Tetranychidae	(Riahi <i>et al.</i> , 2013)
288	<i>Tetranychus urticae</i>	Acari: Tetranychidae	(Bounfour & Tanigoshi, 2001)
289	<i>Tetranychus urticae</i>	Acari: Tetranychidae	(Bayu <i>et al.</i> , 2017)
290	<i>Tetranychus urticae</i>	Acari: Tetranychidae	(Bayu <i>et al.</i> , 2017)
291	<i>Thaumastocoris peregrinus</i>	Hemiptera: Thaumastocoridae	(Barbosa <i>et al.</i> , 2019)
292	<i>Therioaphis maculata</i>	Hemiptera: Aphididae	(Messenger, 1964)
293	<i>Therioaphis maculata</i>	Hemiptera: Aphididae	(Graham, 1959; Messenger, 1964)
294	<i>Therioaphis trifolii</i>	Hemiptera: Aphididae	(Liu <i>et al.</i> , 2012)
295	<i>Thrips hawaiiensis</i>	Thysanoptera: Thripidae	(Cao <i>et al.</i> , 2018)
296	<i>Thrips nigropilosus</i>	Thysanoptera: Thripidae	(Ganaha-Kikumura & Kijima, 2016)
297	<i>Thrips palmi</i>	Thysanoptera: Thripidae	(Kawai & Kawai, 1985)
298	<i>Thrips palmi</i>	Thysanoptera: Thripidae	(Yadav & Chang, 2012)
299	<i>Thrips parvispinus</i>	Thysanoptera: Thripidae	(Sridhar <i>et al.</i> , 2025)
300	<i>Thrips parvispinus</i>	Thysanoptera: Thripidae	(Sridhar <i>et al.</i> , 2025)
301	<i>Thrips tabaci</i>	Thysanoptera: Thripidae	(Murai, 2000)
302	<i>Tirathaba rufivena</i>	Lepidoptera: Pyralidae	(Zhong <i>et al.</i> , 2017)
303	<i>Toxoptera aurantii</i>	Hemiptera: Aphididae	(Wang & Tsai, 2001)
304	<i>Toxoptera citricida</i>	Hemiptera: Aphididae	(Tsai & Wang, 1999)
305	<i>Toxoptera citricida</i>	Hemiptera: Aphididae	(Komazaki, 1982)
306	<i>Toxoptera citricida</i>	Hemiptera: Aphididae	(Komazaki, 1982)
307	<i>Tuta absoluta</i>	Lepidoptera: Gelechiidae	(Campos <i>et al.</i> , 2021)
308	<i>Tuta absoluta</i>	Lepidoptera: Gelechiidae	(Negi <i>et al.</i> , 2020)
309	<i>Tuta absoluta</i>	Lepidoptera: Gelechiidae	(Li <i>et al.</i> , 2019)
310	<i>Tuta absoluta</i>	Lepidoptera: Gelechiidae	(Kanle Satishchandra <i>et al.</i> , 2018)
311	<i>Tuta absoluta</i>	Lepidoptera: Gelechiidae	(Martins <i>et al.</i> , 2016)
312	<i>Tyrophagus similis</i>	Acari: Acaridae	(Kasuga & Amano, 2000)

2. THERMAL PERFORMANCE CURVE MODEL EQUATIONS AND BOOTSTRAPPED SIMULATIONS

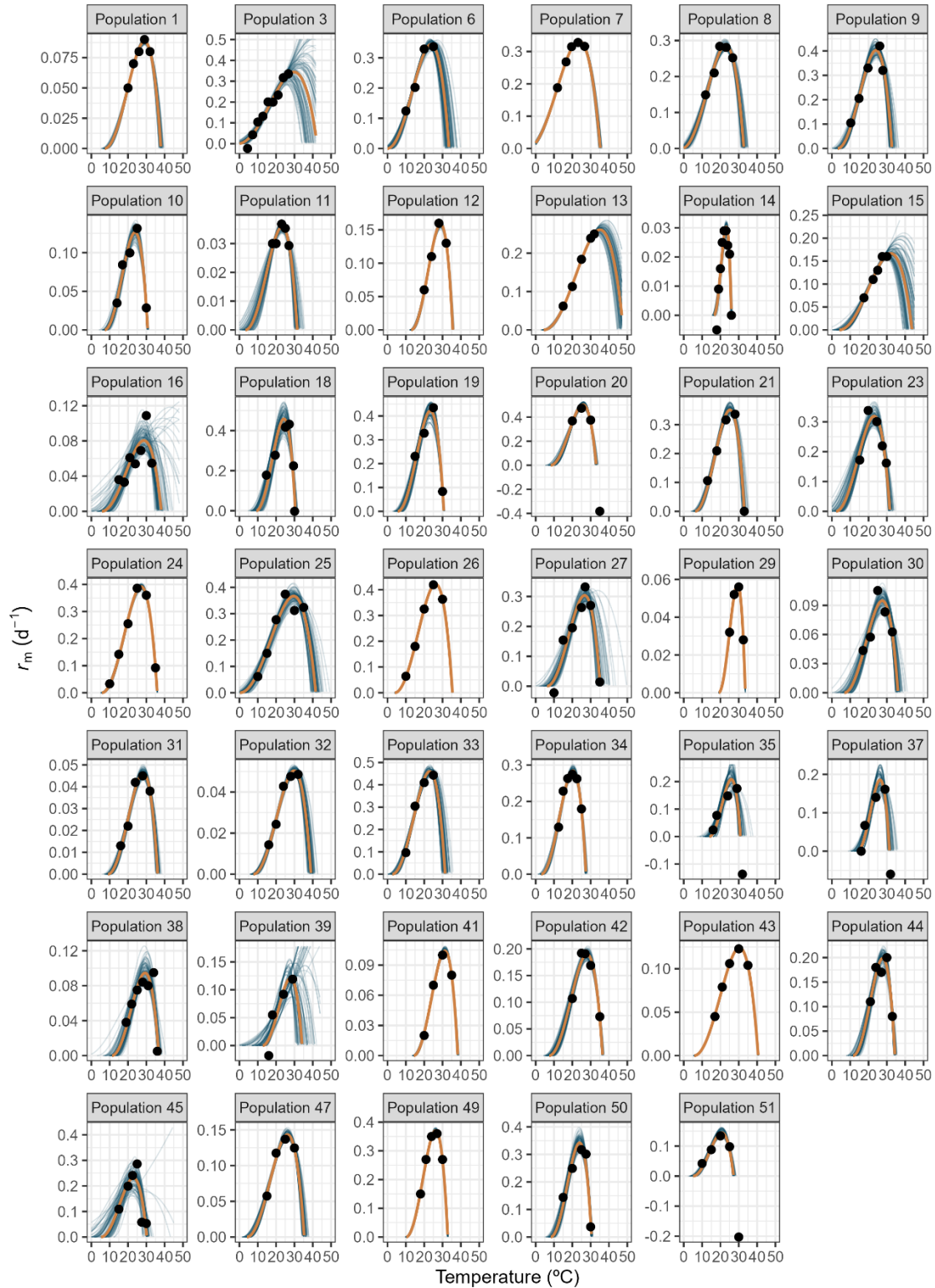
Table S2 describes the 16 TPC model equations used in our approach.

Table S2: TPC equations used to fit nonlinear regression models included in our study.

Model name	Model equation	References
analytis_kontodimas	$r = a \cdot (T - T_{\min})^2 \cdot (T_{\max} - T)$	(Kontodimas et al., 2004)
ashrafi2	$r = a + b \cdot T^{\frac{3}{2}} + c \cdot T^2$	(Ashrafi et al., 2018)
atkin	$r = r_0 \cdot a - b \cdot T^{\frac{T}{10}}$	(Atkin et al., 2005)
beta	$r = \frac{a \left(\frac{T - b + \frac{c}{d+e-2}}{c} \right)^{d-1} \cdot \left(1 - \frac{T - b + \frac{c}{d+e-2}}{c} \right)^{e-1}}{\left(\frac{d-1}{d+e-2} \right)^{d-1} \cdot \left(\frac{e-1}{d+e-2} \right)^{e-1}}$	(Niehaus et al., 2012)
simp_beta	$r = \rho \cdot \left(\alpha - \frac{T}{10} \right) \cdot \left(\frac{T}{10} \right)^{\beta}$	(Damos & Savopoulou-Soultani, 2008)
briere1	$r = a \cdot T \cdot (T - T_{\min}) \cdot (T_{\max} - T)^{\frac{1}{2}}$	(Briere et al., 1999)
simp_briere1	$r = a \cdot (T - T_{\min}) \cdot (T_{\max} - T)^{\frac{1}{2}}$	(Briere et al., 1999)
deutsch	if $T < T_{\text{opt}}$: $r = r_{\max} \cdot \exp\left(-\frac{T - T_{\text{opt}}}{2a}\right)^2$ if $T > T_{\text{opt}}$: $r = r_{\max} \cdot \left(1 - \left(\frac{T - T_{\text{opt}}}{T_{\text{opt}} - CT_{\max}} \right)^2 \right)$	(Deutsch et al., 2008)
jöhnk	$r = r_{\max} \left(1 + a \cdot \left(b^{T - T_{\text{opt}}} - 1 - \frac{\ln b}{\ln c} c^{T - T_{\text{opt}}} - 1 \right) \right)$	(Jöhnk et al., 2008)
pawar	$r = \frac{r_{T_{\text{ref}}} \cdot \exp\left(\frac{-E}{k} \left(\frac{1}{T+273.15} - \frac{1}{T_{\text{ref}}+273.15} \right)\right)}{1 + \left(\frac{E}{E_H - E} \right) \cdot \exp\left(\frac{E_H}{k} \left(\frac{1}{T_{\text{opt}}+273.15} - \frac{1}{T+273.15} \right)\right)}$	(Kontopoulos et al., 2018)
rezende	if $T < b$: $r = a \cdot 10^{\frac{\log_{10} Q_{10}}{\left(\frac{10}{T}\right)}}$ if $T > b$: $r = a \cdot 10^{\frac{\log_{10} Q_{10}}{\left(\frac{10}{T}\right)}} \cdot 1 - c \cdot (b - T)^2$	(Rezende & Bozinovic, 2019)
sharpeschoolfull	$r = \frac{r_{T_{\text{ref}}} \cdot \exp\left(\frac{-E}{k} \left(\frac{1}{T+273.15} - \frac{1}{T_{\text{ref}}+273.15} \right)\right)}{1 + \exp\left(\frac{E_L}{k} \left(\frac{1}{T_L+273.15} - \frac{1}{T+273.15} \right)\right) + \exp\left(\frac{E_H}{k} \left(\frac{1}{T_H+273.15} - \frac{1}{T+273.15} \right)\right)}$	(Schoolfield et al., 1981)
taylor_sexton	$r = r_{\max} \cdot \frac{(T - T_{\min})^4 + 2 \cdot (T - T_{\min})^2 \cdot (T_{\text{opt}} - T_{\min})^2}{(T_{\text{opt}} - T_{\min})^4}$	(Taylor & Sexton, 1972)
thomas	$r = a \cdot \exp^{bT} \left(1 - \left(\frac{T - T_{\text{ref}}}{\frac{c}{2}} \right)^2 \right)$	(Thomas et al., 2012)
tomlinson_phillips	$r = a \cdot (\exp b \cdot T - \exp T - c)$	(Tomlinson & Phillips, 2015)
weibull	$r = a \cdot \left(\frac{c-1}{c} \right)^{\frac{1-c}{c}} \left(\frac{T - T_{\text{opt}}}{b} + \left(\frac{c-1}{c} \right)^{\frac{1}{c}} \right)^{c-1} \cdot \exp\left(-\frac{T - T_{\text{opt}}}{b} + \left(\frac{c-1}{c} \right)^{\frac{1}{c}}\right)^c + \frac{c-1}{c}$	(Angilletta, 2006)

After fitting models with *nls.multstart* package (Padfield et al. 2021), we selected the most suitable TPC following a three-step hierarchical criterion: (1) biological realism—TPCs with unimodal, left-skewed shape, thermal minima above 0°C (below which a positive r_m is unrealistic) and below 42°C (two degrees above the maximum temperature with a positive value in our dataset), respectively; (2) statistical criteria, which was examined using AICs only when there was empirical data along the entire curve, and specifically near the thermal limits ($T_{\min}$, $T_{\max}$ and T_{opt}); whenever the entire curve was not fully represented with empirical data, this goodness-of-fit was not considered; and (3) uncertainty ribbons, which were examined to discriminate among few candidates from previous steps to reinforce the biological criteria (e.g., selecting those with less uncertainty ribbons lying beyond thermal limits).

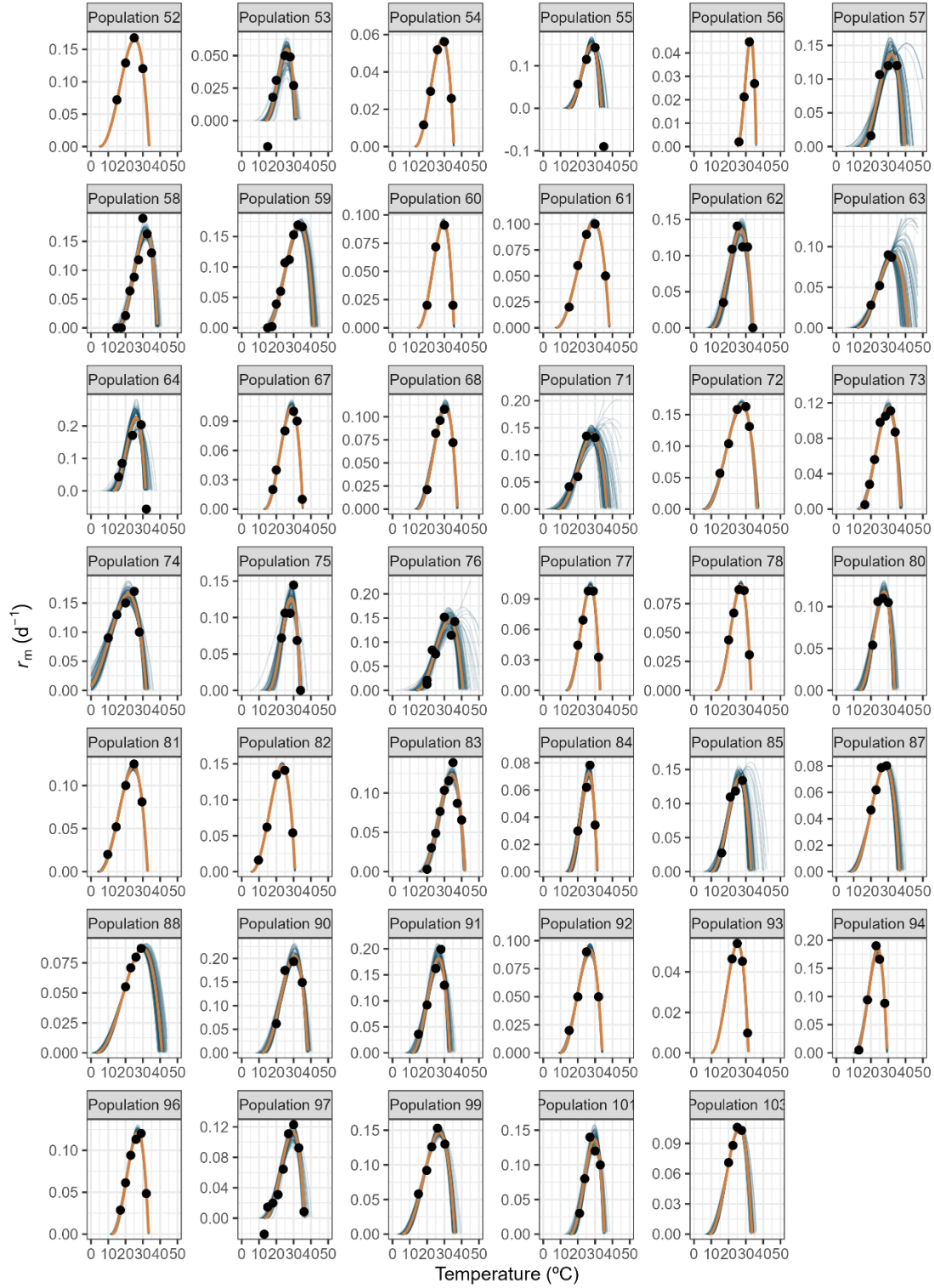
This selection procedure led to 246 curves, each of them with 1,000 bootstrapped curves that serve as the parameter uncertainty ribbons. **Figure S5** depicts all these selected TPCs with their corresponding bootstrapped uncertainties. **Table S3** shows the selected models for each population within studies. The complete three-step procedure is given in *tpcs_selection_filters.csv*. The distribution of these fitted TPCs is shown in **Figure S6**.



Bootstrapping with residual resampling, see 'rTPC' package vignettes

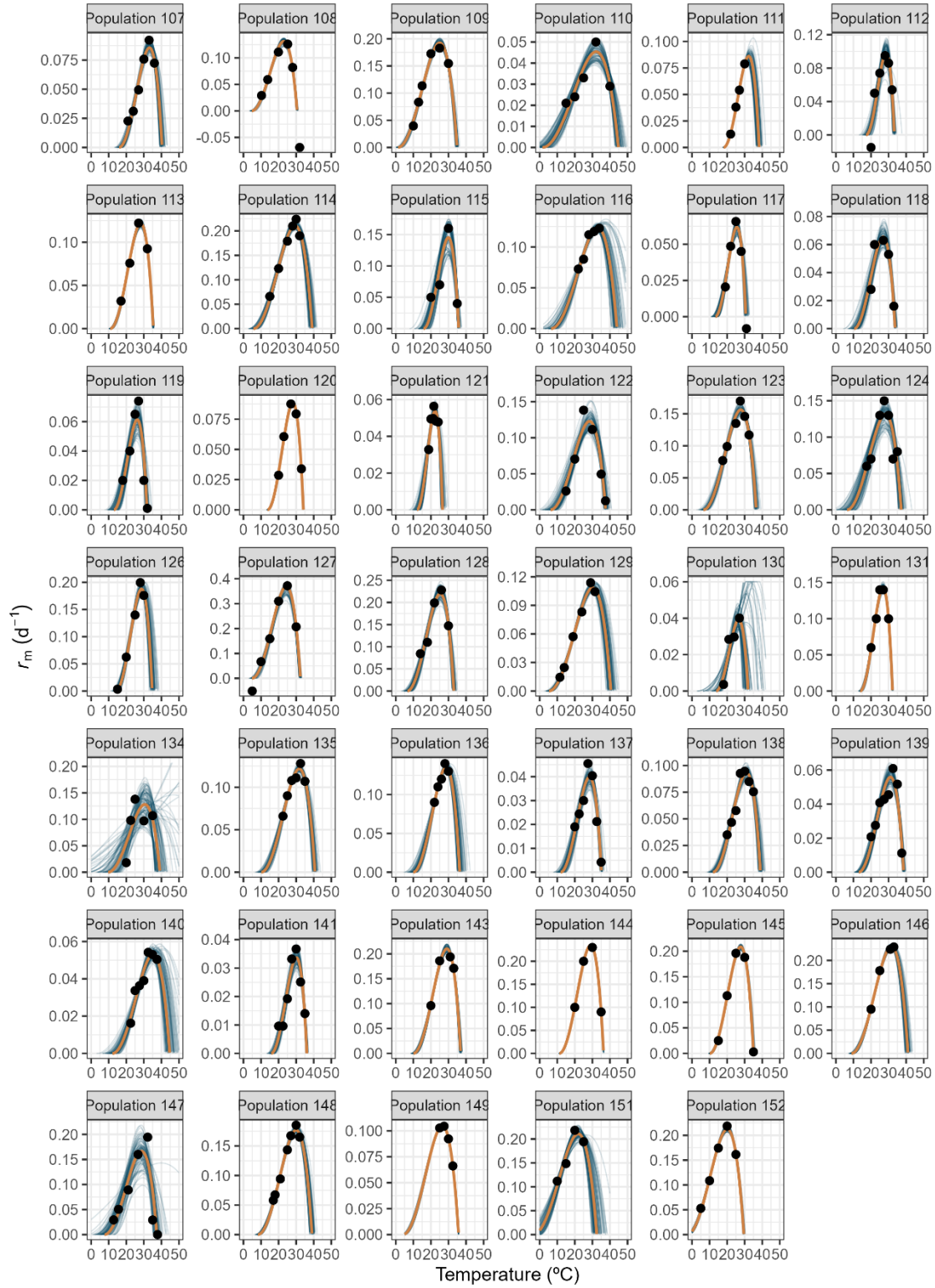
Figure S5: TPC models selected for each population. Points represent the source data, orange line represent the predictions of main model fit and blue lines represent the bootstrapped curves propagating parameter uncertainty. Titles refer to the population id.

Figure S5 (continued)



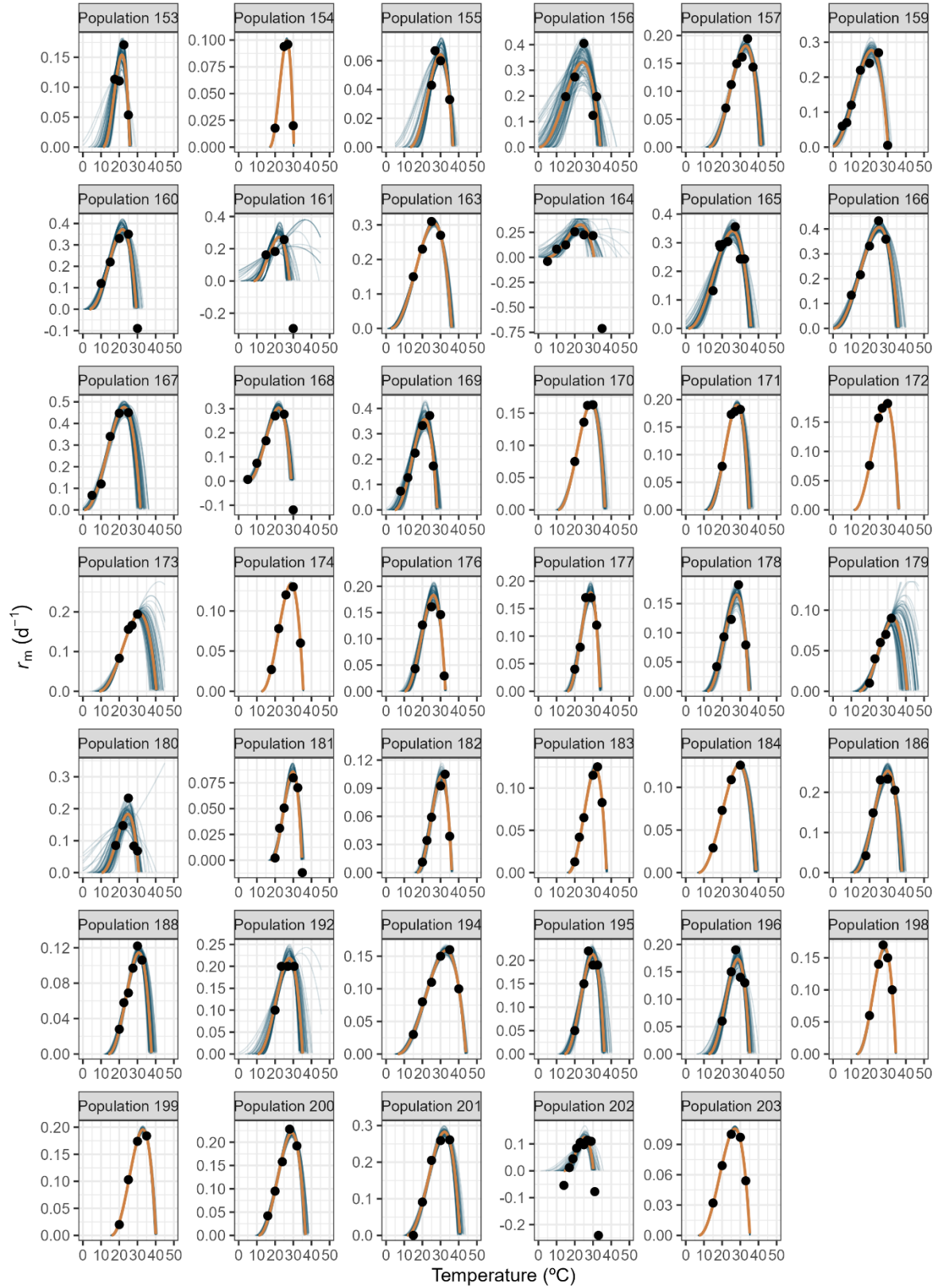
Bootstrapping with residual resampling, see `rTPC` package vignettes

Figure S5 (continued)



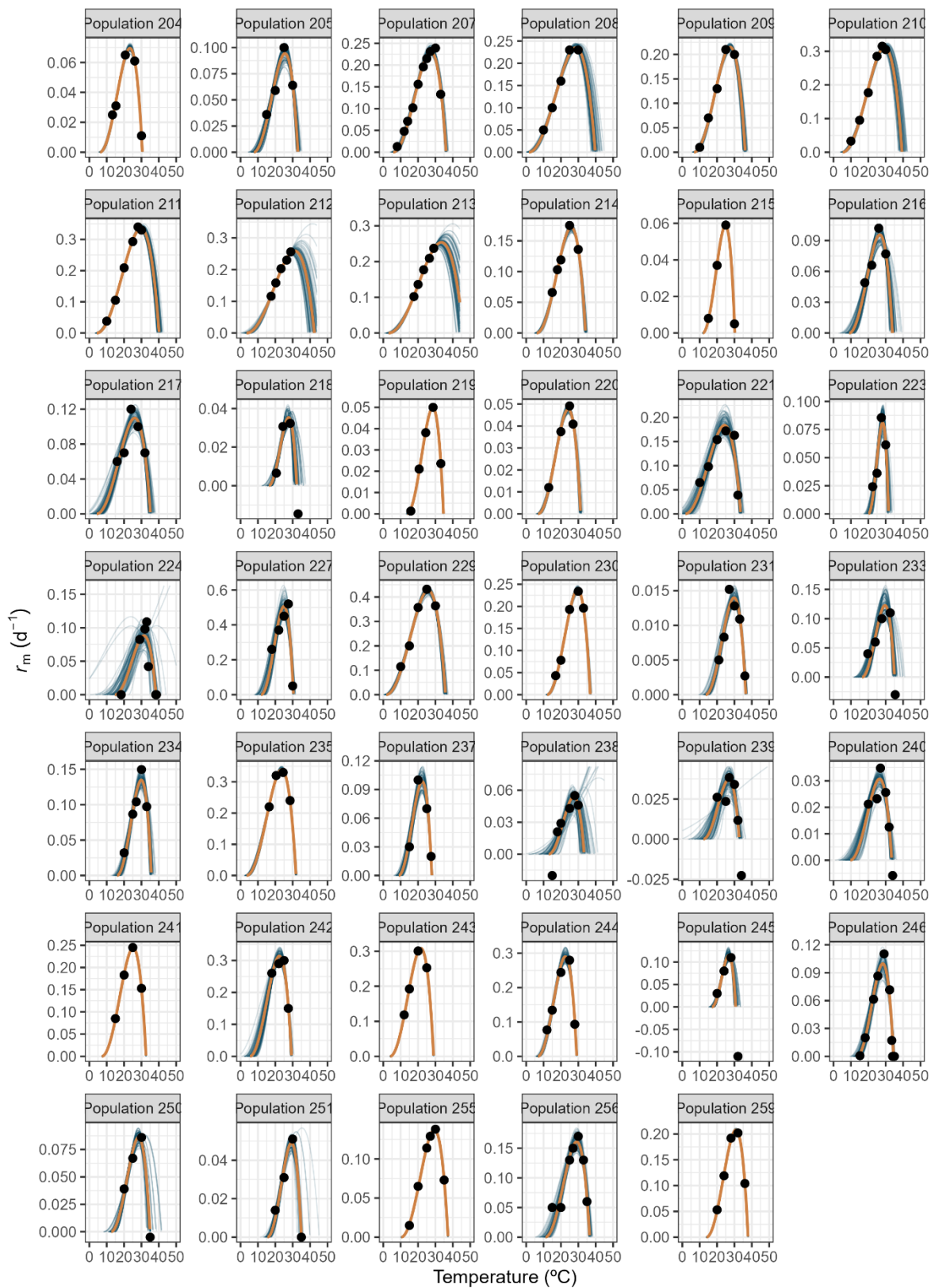
Bootstrapping with residual resampling, see 'rTPC' package vignettes

Figure S5 (continued)



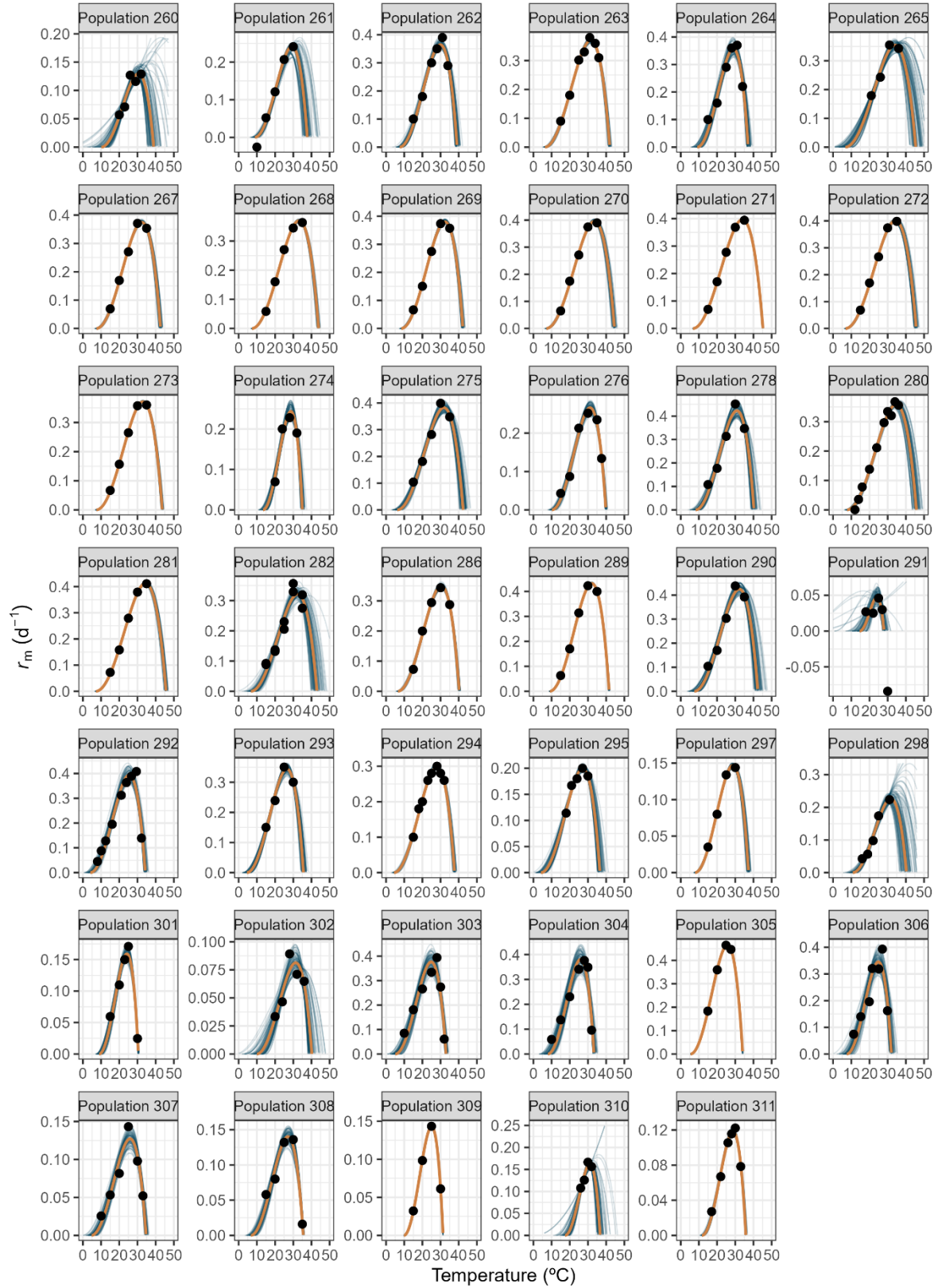
Bootstrapping with residual resampling, see `rTPC` package vignettes

Figure S5 (continued)



Bootstrapping with residual resampling, see 'rTPC' package vignettes

Figure S5 (continued)



Bootstrapping with residual resampling, see `rTPC` package vignettes

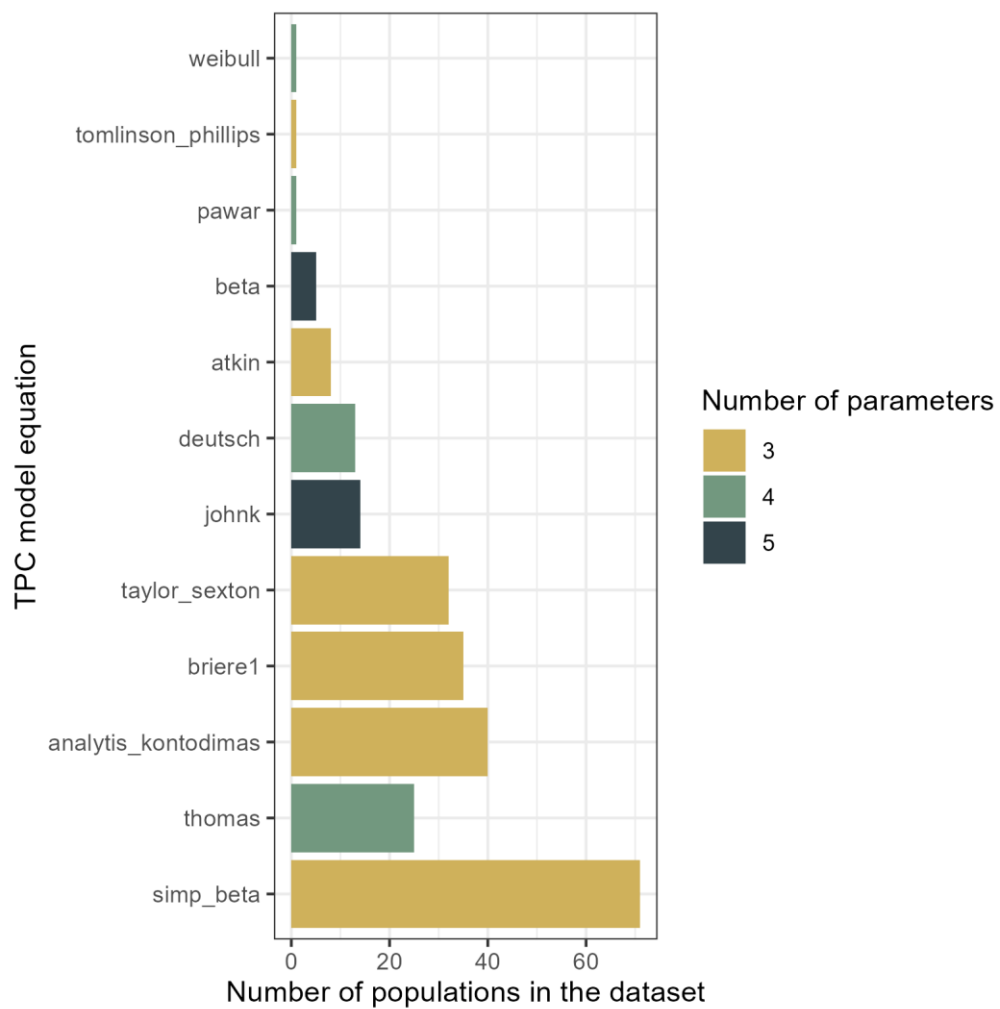


Figure S6: Counts of each TPC model fitted and selected in our dataset (N = 246 fitted TPCs). Colors indicate the number of parameters of the TPC model equation.

Table S3. Selected TPC model equations for each population. Discarded populations due to inability of their models to converge or their unrealistic model fitting are indicated with NA.

ID	Reference	Selected TPC equation
1	yao2016	taylor_sexton
2	gao2013	NA
3	summers1984	johnk
4	rohitha1983	NA
5	rohitha1983	NA
6	lu2008	analytis_kontodimas
7	morgan2001	thomas
8	morgan2001	analytis_kontodimas
9	campbell1977	simp_beta
10	milonas2000	taylor_sexton
11	chi2016	taylor_sexton
12	xia2009	simp_beta
13	jiang2016	thomas
14	azrag2017	thomas
15	pourali2012	briere1
16	toapanta2005	simp_beta
17	algerrawy2020	NA
18	komazaki1982	johnk
19	akca2015	simp_beta
20	mccornack2004	simp_beta
21	tian2020	johnk
22	kocourek1994	NA
23	komazaki1982	analytis_kontodimas
24	xia1999	taylor_sexton
25	parajulee2007	briere1
26	zamani2006	simp_beta
27	wang1997	simp_beta
28	eyidozehi2023	NA
29	basirat2015	atkin
30	guo2017	analytis_kontodimas
31	li2024	taylor_sexton
32	li2020	simp_beta
33	jandricic2010	analytis_kontodimas
34	lee2008	thomas

ID	Reference	Selected TPC equation
35	vargas1997	thomas
36	yang1994a	NA
37	vargas1997	NA
38	yang1994b	beta
39	vargas1997	simp_beta
40	jeon2012	NA
41	choudhary2020	briere1
42	wang1996	analytis_kontodimas
43	bonato2007	johnk
44	ma2017	taylor_sexton
45	satar2002	atkin
46	yazdanpanah2019	NA
47	li2015	analytis_kontodimas
48	soh2018	NA
49	kim2014	thomas
50	satar2005	simp_beta
51	deloach1974	simp_beta
52	ullah2020	briere1
53	khadioli2014b	thomas
54	legaspi2007	weibull
55	lin2023	simp_beta
56	ren2016	taylor_sexton
57	giga1983	taylor_sexton
58	daglish2021	thomas
59	daglish2021	johnk
60	giga1983	analytis_kontodimas
61	elaalaoui2024a	thomas
62	walton2010	analytis_kontodimas
63	bazgir2015	simp_beta
64	vargas1997	simp_beta
65	mbapila2002	NA
66	mbapila2002	NA
67	khadioli2014a	johnk
68	singh1991	analytis_kontodimas
69	mbapila2002	NA
70	mbapila2002	NA
71	delpino2020	simp_beta

ID	Reference	Selected TPC equation
72	pasqualotto2024	simp_beta
73	tokuda2005	analytis_kontodimas
74	durak2013	briere1
75	dreyer1996	simp_beta
76	dreyer1996	briere1
77	cheng1988	simp_beta
78	chang1988	thomas
79	chang1988	NA
80	pang2022	analytis_kontodimas
81	gould2005	thomas
82	gould2005	deutsch
83	smith1965	analytis_kontodimas
84	aghda,2009	simp_beta
85	granett1987	analytis_kontodimas
86	madden1986	NA
87	madden1986	taylor_sexton
88	madden1986	briere1
89	yang2024	NA
90	hosseinzade2015	analytis_kontodimas
91	liu2000	deutsch
92	elaalaoui2024b	simp_beta
93	rodriguez1989	taylor_sexton
94	schlesener2020	analytis_kotnodimas
95	clarke1959	NA
96	qin2013	beta
97	sandhu2013	atkin
98	bounfour2001	NA
99	bonato1990	simp_beta
100	wang2024	NA
101	wang2024	analytis_kontodimas
102	jafarian2016	NA
103	siddiqui1973	simp_beta
104	imura1986	NA
105	imura1986	NA
106	imura1986	NA
107	liu2018	thomas
108	danthanarayana1995	simp_beta

ID	Reference	Selected TPC equation
109	asante1991	analytis_kontodimas
110	elaalaoui2025	briere1
111	iranipour2010	taylor_sexton
112	rodrigues2023	analytis_kontodimas
113	lin2020	simp_beta
114	childers1991	johnk
115	lowry1992	simp_beta
116	zheng2008	briere1
117	chen2019	taylor_sexton
118	mermer2023	analytis_kontodeimas
119	mermer2023	pawar
120	govindan2020	analytis_kontodimas
121	dalal2021	taylor_sexton
122	noorulane2018	deutsch
123	mironidis2014	taylor_sexton
124	mironidis2008	analytis_kontodimas
125	mironidis2008	NA
126	ramalho2015	simp_beta
127	deloach1974	thomas
128	latham2011	simp_beta
129	zahiri2010	simp_beta
130	wei2023	analytis_kontodimas
131	jaramillo2009	taylor_sexton
132	he2014	NA
133	wang2021	NA
134	howe1957	taylor_sexton
135	opit2010	briere1
136	islam2020	simp_beta
137	jiang2008	atkin
138	wang2000	taylor_sexton
139	tang2008	simp_beta
140	wang2009	simp_beta
141	dong2007	atkin
142	hassan2011	NA
143	wang2014	analytis_kontodimas
144	he1999	analytis_kontodimas
145	haghani2007	analytis_kontodimas

ID	Reference	Selected TPC equation
146	wang2014	taylor_sexton
147	strong1970	simp_beta
148	ugine2012	simp_beta
149	zamanzandeh2022	analytis_kontodimas
150	lei2021	NA
151	zhou2014	briere1
152	barlow1962	briere1
153	olmez2003	simp_beta
154	liao2017	analytis_kontodimas
155	kobori2017	taylor_sexton
156	souza2020	analytis_kontodimas
157	sanchezramos2017	deutsch
158	qin2017	NA
159	ahn2025	briere1
160	ahn2025	simp_beta
161	li2025	taylor_sexton
162	zhou2014	NA
163	barbosa2011	simp_beta
164	davis2006	johnk
165	davis2006	analytis_kontodimas
166	eldin1976	brierre1
167	barlow1962	thomas
168	deloach1974	thomas
169	diaz2005	johnk
170	valle1986	simp_beta
171	valle1986	analytis_kontodimas
172	valle1986	analytis_kontodimas
173	valle1986	simp_beta
174	sahoo2025	thomas
175	wang2024	NA
176	chen2016	briere1
177	quan2019	deutsch
178	lin2013	deutsch
179	lu2016	briere1
180	fonseca2019	atkin
181	beckett1994	thomas
182	beckett1994	beta

ID	Reference	Selected TPC equation
183	beckett1994	johnk
184	scheunemann2021	taylor_sexton
185	scheunemann2021	NA
186	han2007	analytis_kontodimas
187	chen2017	NA
188	park2017	simp_beta
189	nagaraju2024	NA
190	wang2023	NA
191	varikou2012	NA
192	lema1985	analytis_kontodimas
193	harbi2025	NA
194	fand2014	briere1
195	chen2015	taylor_sexton
196	chen2015	analytis_kontodimas
197	kumar2013	NA
198	chen2015	atkin
199	kumar2013	briere1
200	golizadeh20102	thomas
201	elekçioglu2004	briere1
202	allen1995	taylor_sexton
203	amiri2010	thomas
204	geng2017	taylor_sexton
205	satar2017	simp_beta
206	saeed2019	NA
207	dan1995(in shi2012)	beta
208	marchioro2012	briere1
209	ngowi2017	thomas
210	golizadeh2009	thomas
211	golizadeh2009	simp_beta
212	sarnthoy1989	briere1
213	sarnthoy1989	briere1
214	ko2003	simp_beta
215	erkiliç1997	briere1
216	heidari1999	simp_beta
217	zhang2024	taylor_sexton
218	allsopp1981	taylor_sexton
219	allsopp1981	thomas

ID	Reference	Selected TPC equation
220	howe1953	simp_beta
221	pasqualotto2024	briere1
222	fidelis2019	NA
223	mejia2019	tomlinson_phillips
224	birch1953	atkin
225	birch1953	NA
226	birch1953	NA
227	asin2001	deutsch
228	auad2009	NA
229	tsai1998	simp_beta
230	bahirai2019	taylor_sexton
231	peng2016	deutsch
232	tian2022	NA
233	ahn2019	thomas
234	rae1991	deutsch
235	pendleton2009	analytis_kontodimas
236	pendleton2009	NA
237	hoddle2002	analytis_kontodimas
238	khadioli2014b	taylor_sexton
239	arbabtafti2021	simp_beta
240	arbabtafti2021	simp_beta
241	taylor2023	taylor_sexton
242	asin2001	simp_beta
243	turak1998	simp_beta
244	turak1998	deutsch
245	lu2024	simp_beta
246	birch1953	taylor_sexton
247	birch1953	NA
248	birch1953	NA
249	birch1953	NA
250	hansen2004	simp_beta
251	hansen2004	simp_beta
252	mobarak2022	NA
253	tata2025	NA
254	ouda2022	NA
255	prasad2022	johnk
256	rao2020	simp_beta

ID	Reference	Selected TPC equation
257	rao2014	NA
258	manimanjari2014	NA
259	martinez2014	deutsch
260	aysal2008	simp_beta
261	kuo2006	briere1
262	he2008	johnk
263	he2008	briere1
264	he2008	simp_beta
265	bonato1999	briere1
266	demoraes1987	NA
267	gotoh2010	simp_beta
268	gotoh2010	briere1
269	gotoh2010	deutsch
270	gotoh2010	briere1
271	gotoh2010	briere1
272	gotoh2010	briere1
273	gotoh2010	thomas
274	veerlapati2023	analytis_kontodimas
275	ullah2011	simp_beta
276	hasanvand2020	simp_beta
277	ristyadi2021	NA
278	gotoh2015 (in nachman2015)	simp_beta
279	tanigoshi1975 (in logan1976)	NA
280	roy2003	briere1
281	ullah2011	johnk
282	gotoh2015 (in nachman2015)	simp_beta
283	gotoh2004	NA
284	gotoh2015 (in nachman2015)	NA
285	karami2012	NA
286	gotoh2014 (in nachman2015)	thomas
287	riahi2013	NA
288	bounfour2001	NA
289	bayu2017	simp_beta
290	bayu2017	simp_beta
291	barbosa2019	simp_beta
292	messenger1964	beta
293	graham1959 (in messenger1964)	simp_beta

ID	Reference	Selected TPC equation
294	liu2012	analytis_kontodimas
295	cao2018	taylor_sexton
296	kikumura2016	NA
297	kawai1985	simp_beta
298	yadav2012	simp_beta
299	sridhar2025	NA
300	sridhar2025	NA
301	murai2000	simp_beta
302	zhong2017	taylor_sexton
303	wang2001	briere1
304	tsai1999	johnk
305	komazaki1982	simp_beta
306	komazaki1982	simp_beta
307	decampos2021	simp_beta
308	negi2020	briere1
309	li2019	briere1
310	satishchandra2018	simp_beta
311	martins2016	deutsch

3. CLIMATIC DATA

Figure S7 shows the distribution of historic and future climate data set across source studies locations for monthly temporal resolution.

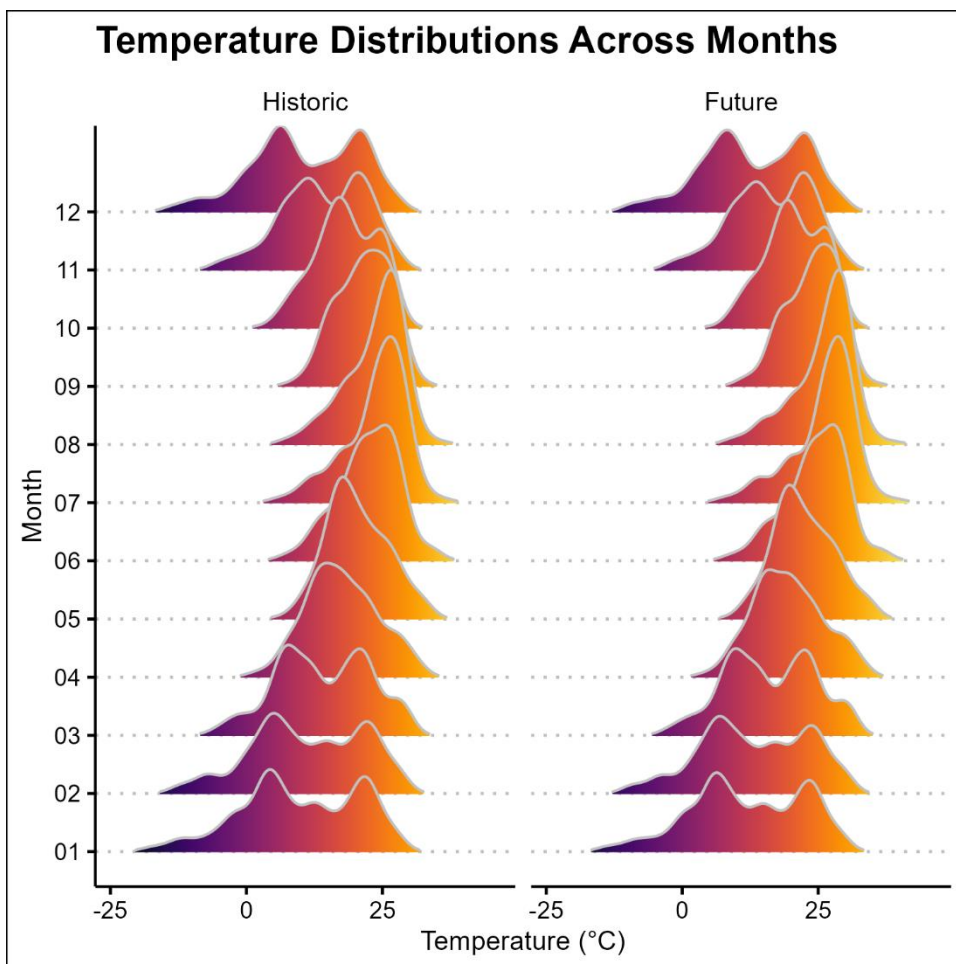


Figure S7: density graphs with the distribution of monthly temperatures pooled across source locations at both Historic and Future climatic datasets.

Figure S8 shows the variability across CMIP6 models for this temperature in 2050.

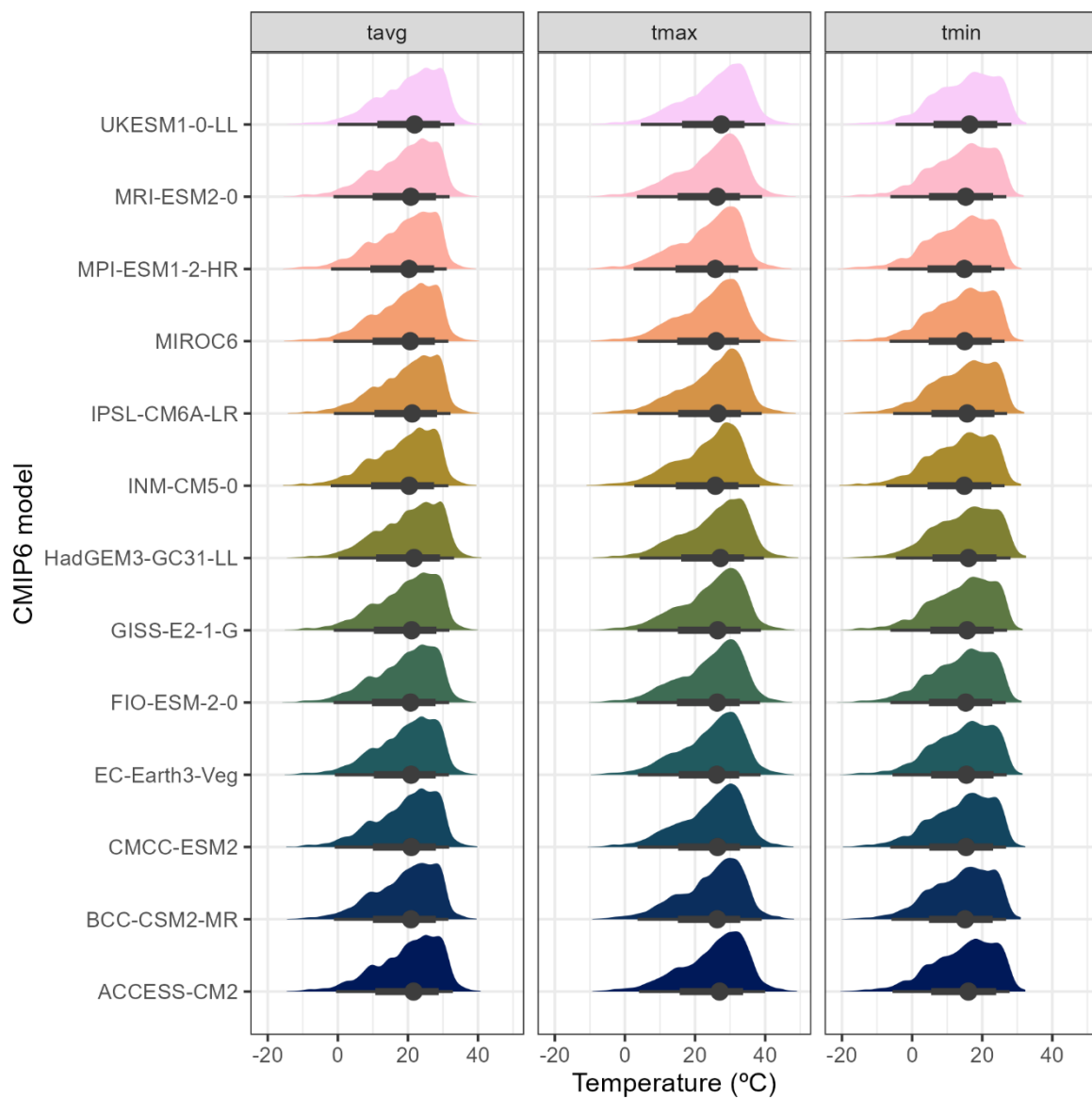


Figure S8: density plots showing the distribution of monthly temperatures (average, or *tavg*; maximum, or *tmax*; and minimum, or *tmin*) for each CMIP6 model (each model has its own colour).

4. THERMAL LIMITS VARIATION

Table S4 shows the estimates and standard error of thermal minimum, maximum and optimum according to varying-intercept hierarchical models. Additionally, Table S3 shows, according to hierarchical models using either order or absolute latitude as predictors, significant differences of thermal limits estimates across taxonomic orders for both thermal limits and only a significant negative correlation between thermal minimum and latitude.

In Figure 1 of the main text, the high isolated point of $T_{\min}$ (25.14°C) corresponds to the locust *Calliptamus italicus* (Ren et al. 2016; id 56). This extremely high value is given by an extremely low experimental value of r_m at a mild temperature (26°C). However, this data point seems to agree with previous empirical data in the literature, as Ren et al. (2016) report no egg-laying of females at 23°C according to previous experimental studies, which may reflect a rare and extraordinary lower thermal limit for populations of *C. italicus*. Rather than invalidating the modelling framework, this case illustrates the mechanisms why TPCs for population-level rates are typically narrower than those for organism-level rates. The 0°C values of $T_{\min}$ inform about populations with few, but low rate data in the cold-end of the TPC fitted with *briere1* equation, which is the only one that estimates positive $T_{\min}$ values; thus, these estimates might be slightly biased towards colder temperatures.

Table S4: Parameters and p -values of the hierarchical models for the three thermal limits.

Thermal trait	Intercept-only model	Intercept and slope models		
	Estimate $\pm$ s.e.	Latitude group (p -value)	Order (p -value)	Latitude $\bar{\beta}_{\text{latitude}}(\text{SE}) + \alpha$ p -value
$T_{\min}$	10.06 ± 0.34	$p < 0.001$,	$p < 0.001$,	$-0.04(0.02) + 11.27$
	$\sigma_{\text{species}} = 3.56$	$\sigma_{\text{species}} = 3.16$,	$\sigma_{\text{species}} = 3.14$,	$\left(p = 0.04 \right)$
	$\sigma_{\text{study}} = 2.33$	$\sigma_{\text{study}} = 2.62$	$\sigma_{\text{study}} = 2.47$	$\left(\sigma_{\text{species}} = 3.44, \sigma_{\text{study}}^2 = 2.35 \right)$
$T_{\max}$	35.18 ± 0.27	$p < 0.05$,	$p < 0.005$,	$-0.01(0.01) + 35.54$
	$\sigma_{\text{species}} = 2.02$	$\sigma_{\text{species}} = 1.94$,	$\sigma_{\text{species}} = 1.94$,	$\left(p = 0.37 \right)$
	$\sigma_{\text{study}} = 2.80$	$\sigma_{\text{study}} = 2.83$	$\sigma_{\text{study}} = 2.68$	$\left(\sigma_{\text{species}} = 2.01, \sigma_{\text{study}} = 2.83 \right)$
T_{opt}	27.77 ± 0.22 ,	$p < 0.001$,	$p < 0.001$	$-0.01(0.01) + 28.11$
	$\sigma_{\text{species}} = 1.81$,	$\sigma_{\text{species}} = 3.16$,	$\sigma_{\text{species}} = 2.55$,	$\left(p = 0.26, \right)$
	$\sigma_{\text{study}} = 2.17$	$\sigma_{\text{study}} = 2.62$	$\sigma_{\text{study}} = 2.25$	$\left(\sigma_{\text{species}} = 1.78, \sigma_{\text{study}} = 2.19 \right)$

Figure S9 shows the distribution of $T_{\min}$ and $T_{\max}$ for low- and high-latitude populations. Figure S10 shows the latitudinal variation of cold- and warm safety margins using yearly average temperatures as experienced temperature by the populations at their source locations (i.e., T_{amb}). Alternatively, in Figure S11 we used the temperature experienced at the three coldest and the three warmest months at the source locations as T_{amb} for calculating cold- and warm safety margins, respectively.

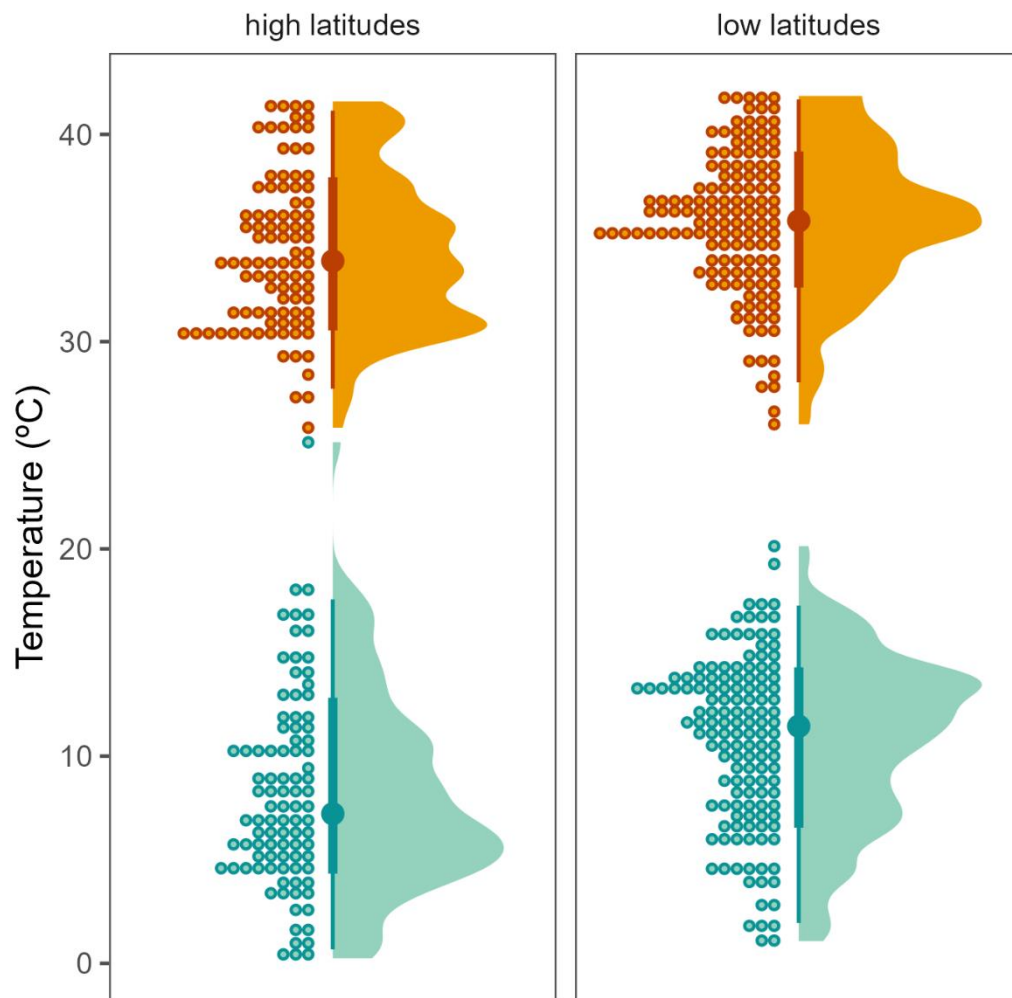


Figure S9. Rain plots showing the distribution of estimates of thermal limits for populations from high latitudes (left) and low latitude (right).

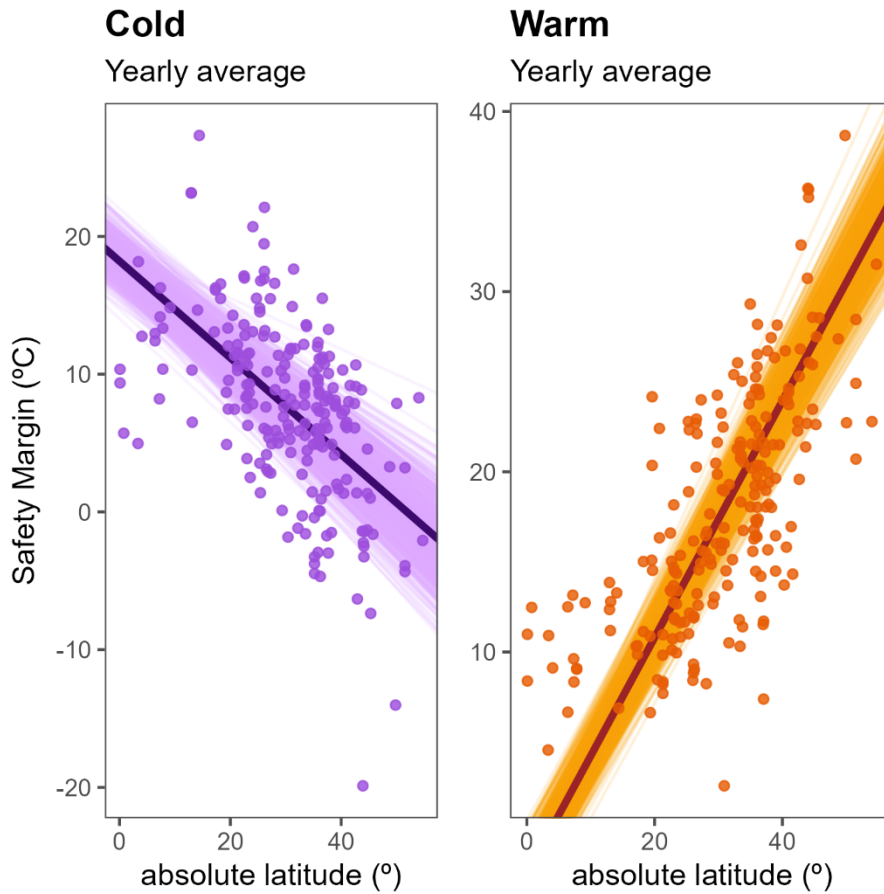


Figure S10. Linear hierarchical models describing the latitudinal variation of cold (left, purple) and warm (right, orange) safety margins. Cold and warm safety margins are defined as the difference between experienced temperature and either $T_{\min}$ or $T_{\max}$, respectively. Here, yearly average temperature at the source locations is used to measure the experienced temperature at the source locations. Models were fitted with *rma.v()* function from the *metafor* R package (Viechtbauer, 2010), with *study* and *reference* as grouping variables and the inverse of the bootstrapped variances of thermal limits as weights. Dark lines represent the predictions obtained from the fitted linear hierarchical model (multivariate meta-analysis), whereas each light line represents the predictions of each bootstrapped new simulated model (see Methods). Points represent the estimates at each source latitude.

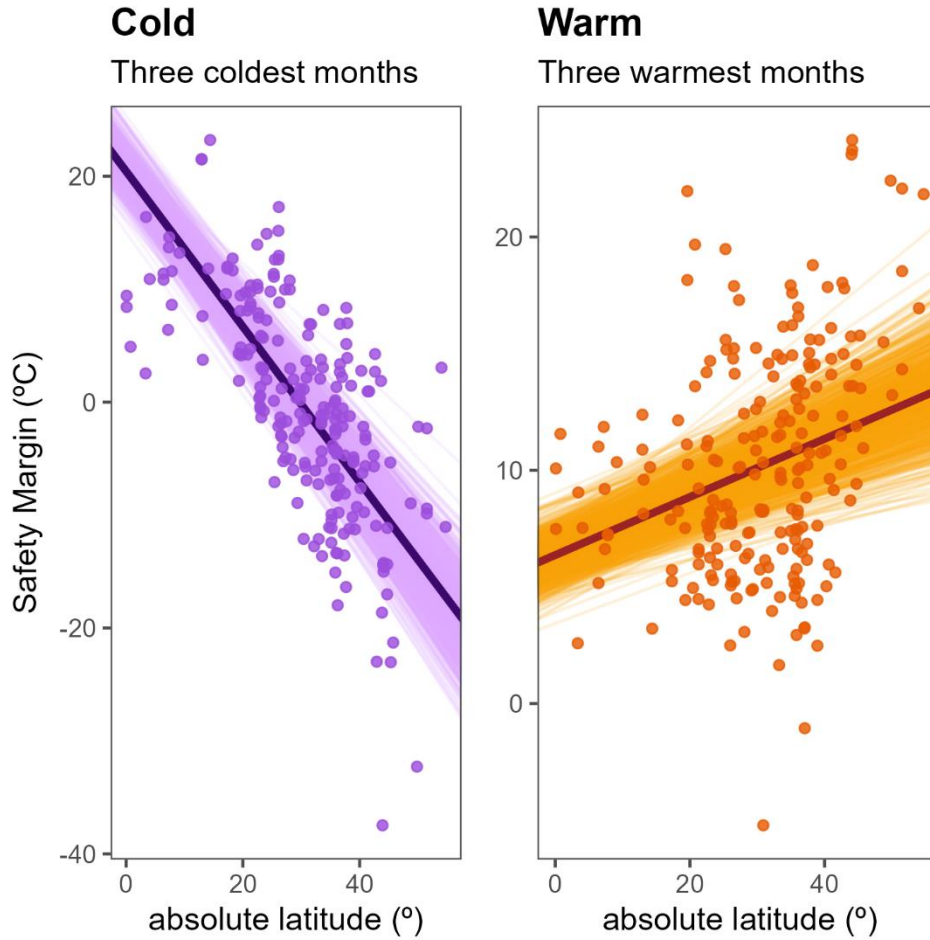


Figure S11. Linear hierarchical models describing the latitudinal variation of cold (left, purple) and warm (right, orange) safety margins. Cold and warm safety margins are defined as the difference between experienced temperature and either $T_{\min}$ or $T_{\max}$, respectively. Here, yearly average temperature at the source locations is used to measure the experienced temperature at the source locations. Models were fitted with *rma.v()* function from the *metafor* R package (Viechtbauer, 2010), with *study* and *reference* as grouping variables and the inverse of the bootstrapped variances of thermal limits as weights. Dark lines represent the predictions obtained from the fitted linear hierarchical model (multivariate meta-analysis), whereas each light line represents the predictions of each bootstrapped new simulated model (see Methods). Points represent the estimates at each source latitude.

5. PERFORMANCE SHIFT INDEX (Ψ_r)

An interactive map in **.html* format is given in the GitHub repository website (LINK HERE) including the information illustrated in Figure 2 of the main text.

The distribution of Ψ_r was left-skewed (mean = 0.064; median = 0.075; Figure S12).

After averaging Ψ_r across months, we found no significant effect of absolute latitude on Ψ_r ($F_{7, 165.21} = 1.40$, $p = 0.21$) (Figure S13). We also found no significant effect of taxonomic order on Performance Shift Index (Ψ_r) ($t_{84.34} = 1.54$, $p = 0.13$, ANOVA Type III).

We also visually examined the variation of Ψ_r across latitudes in both hemispheres (i.e., with raw rather than absolute latitude), both for the yearly average values of Ψ_r (Figure S14) and for each month (Figure S15).

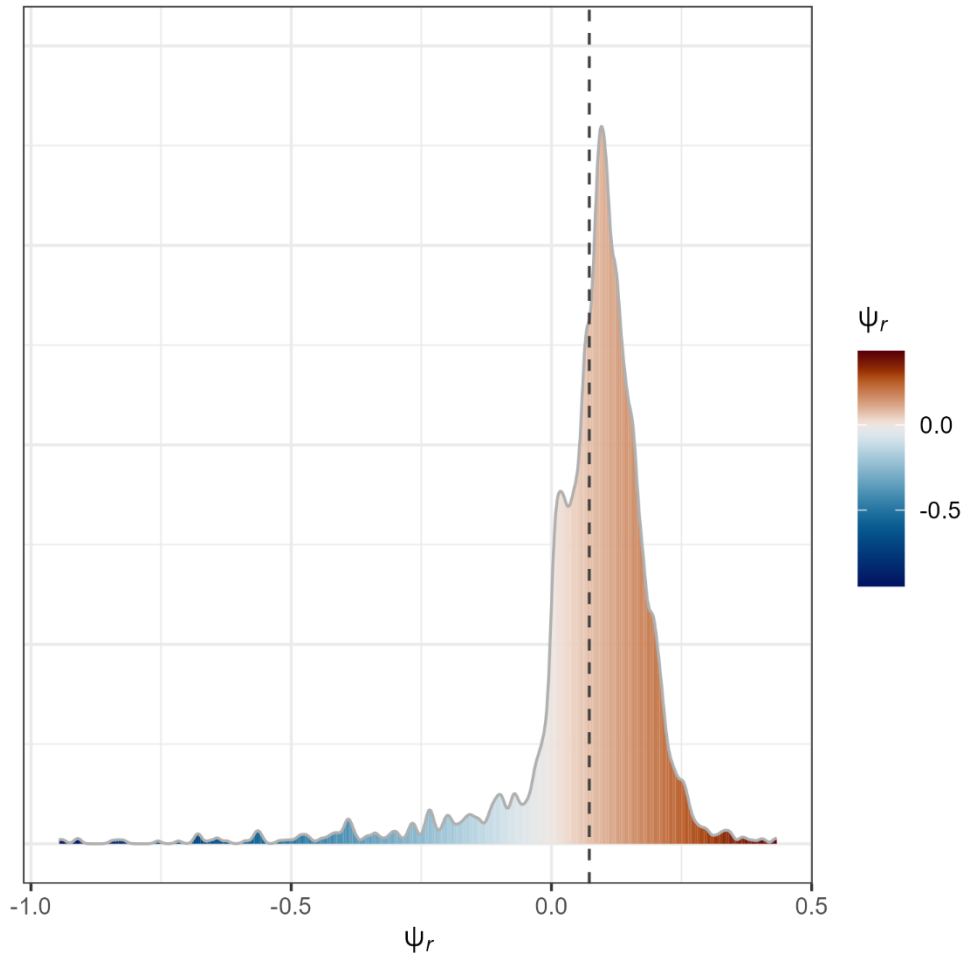


Figure S12: Distribution of Ψ_r . Density plot showing the distribution of Ψ_r of all populations under study after excluding months with predicted $r_m = 0$ at both historic and future climates due to temperatures outside the thermal tolerance range of the fitted TPC (i.e., below $T_{\min}$ or above $T_{\max}$). The dashed line represents the mean value of Ψ_r across populations.

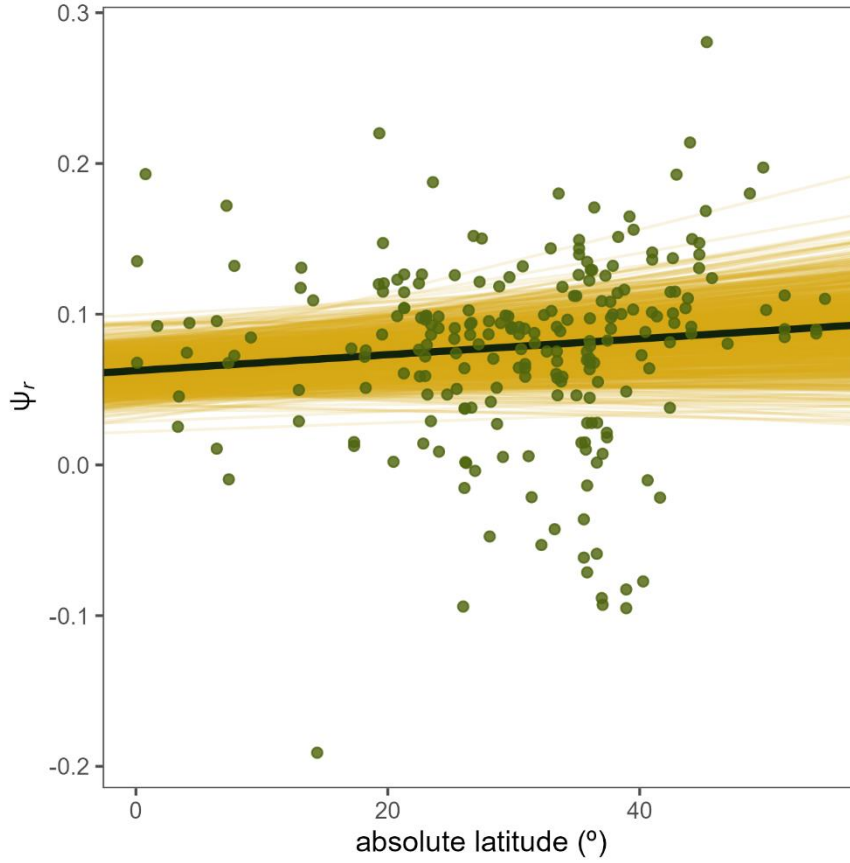


Figure S13: Model predictions of the effects of absolute latitude on Ψ_r . Predictions from the linear hierarchical model of Ψ_r , using absolute latitude as a covariate and study and species as grouping variables. Previously to model fitting, months with predicted $r_m = 0$ at both historic and future climates due to temperatures outside the thermal tolerance range of the fitted TPC (i.e., below $T_{\min}$ or above $T_{\max}$) were excluded. The dark line represents the predictions obtained from the fitted linear hierarchical, and light lines represent the simulated regression lines obtained by sampling slope and intercept from their estimated normal distributions (based on standard errors). Points represent the Ψ_r estimates at each source latitude.

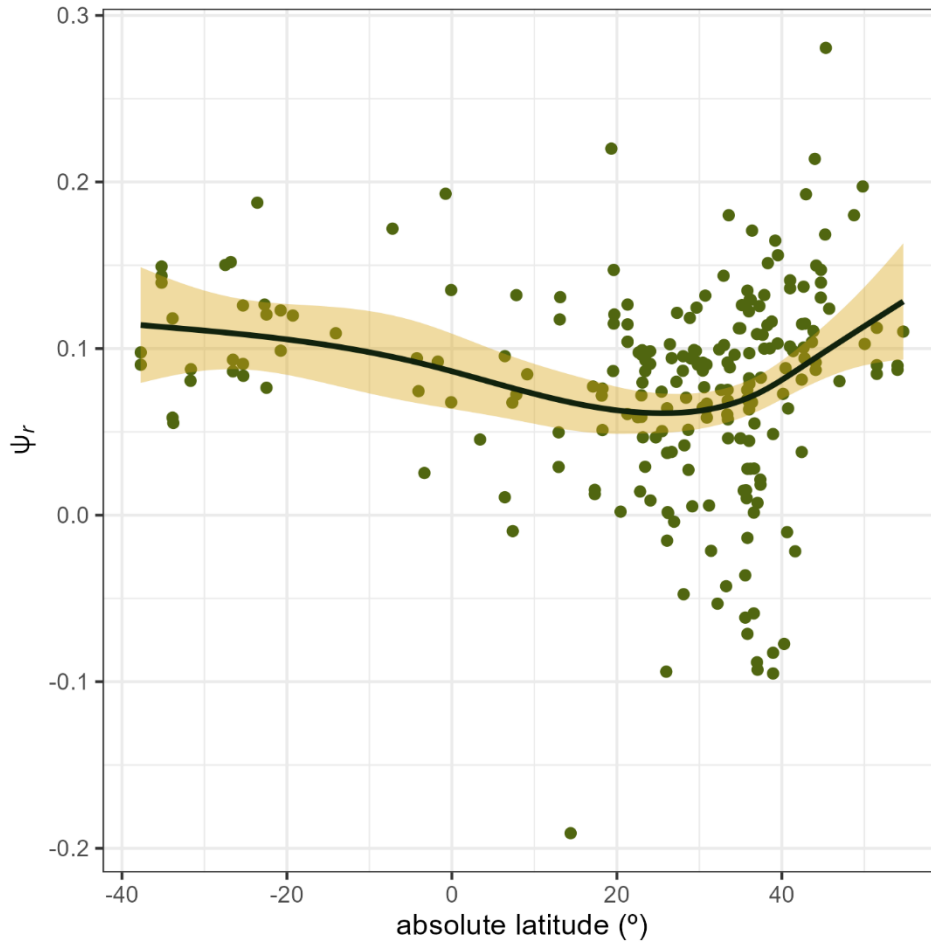


Figure S14: Latitudinal variation of Ψ_r at both hemispheres. Plot shows the variation of Ψ_r across latitudes at both hemispheres, where negative latitude values correspond to the Southern Hemisphere and positive ones to the Northern Hemisphere. The dark line represents a smooth line fitted with a *loess* regression model, and points represent each yearly average value of Ψ_r .

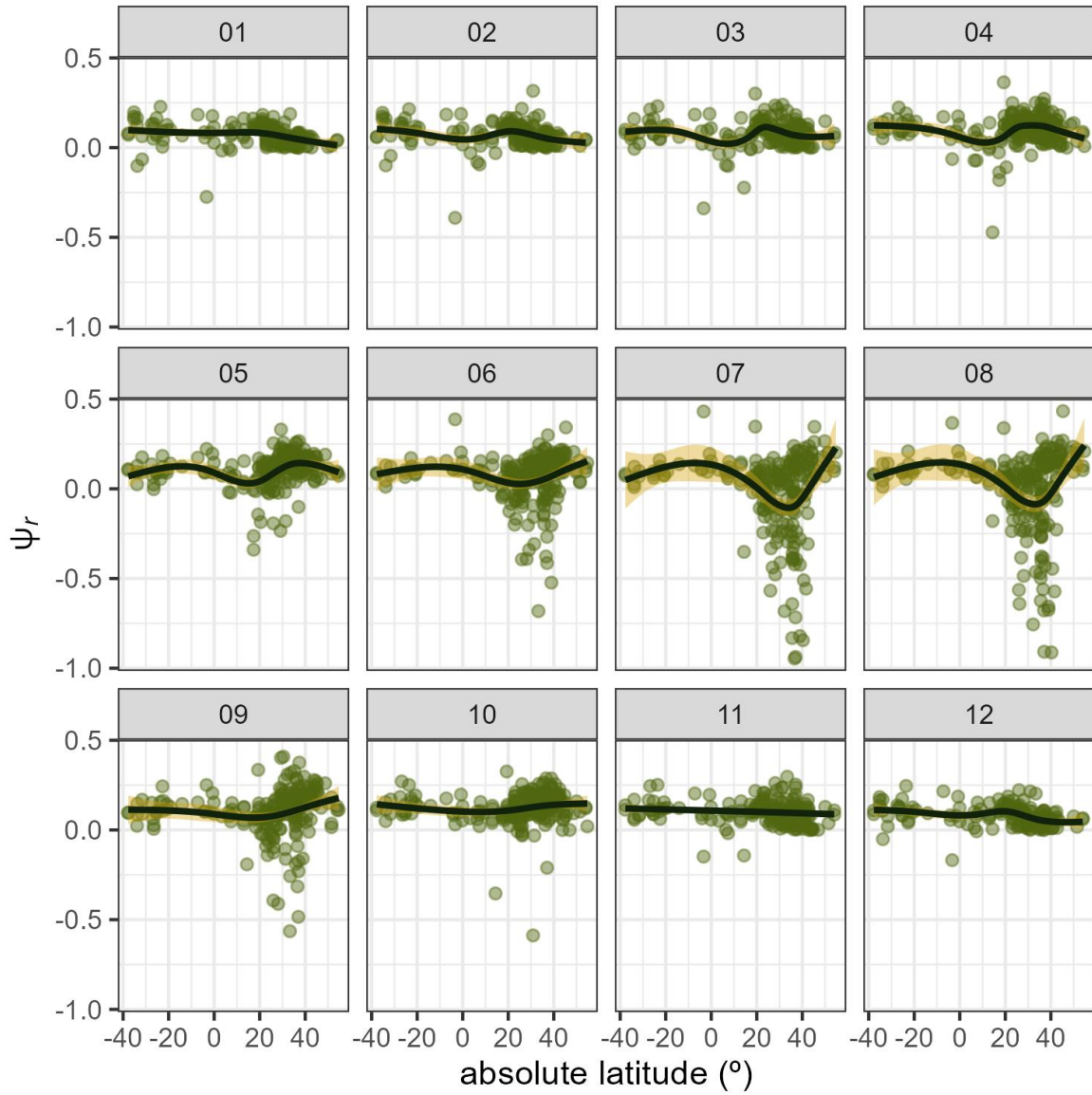


Figure S15: Latitudinal variation of Ψ_r at both hemispheres across months. Plots show the variation of Ψ_r across latitudes at both hemispheres, where negative latitude values correspond to the Southern Hemisphere and positive ones to the Northern Hemisphere. Each panel corresponds to a month, chronologically ordered. The dark lines represent smooth lines fitted with loess regression models, and points represent each monthly average value of Ψ_r .

6. PROJECTING PERFORMANCE OVER TPC ZONES

Figure S16 shows the proportion of TPC zones potentially occupied by populations at their source locations under historical and future climatic data, respectively.

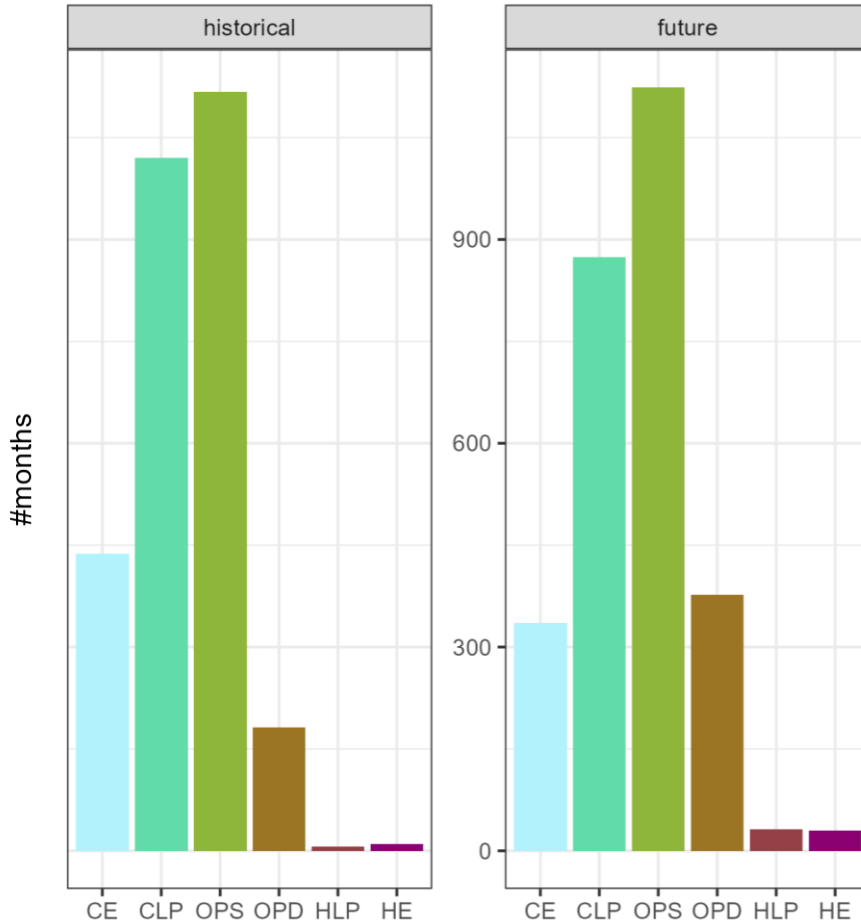


Figure S16: total number of months per year placed within each population-specific TPC region, using monthly temperatures and pooled for all populations. These TPC regions and their colours are defined as follows: *Cold Exclusion* (light blue, CE , $T_{\text{month}} < T_{\text{min}}$), *Cold-Limited Performance* (turquoise, CLP , $T_{\text{min}} \leq T_{\text{month}} < T_{50L}$), *Optimal Performance & Safe* (green, OPS , $T_{50L} \leq T_{\text{month}} < T_{\text{opt}}$), *Optimal Performance & Danger* (brown, OPD , $T_{\text{opt}} \leq T_{\text{month}} < T_{50R}$), *Heat-Limited Performance* (clay, HLP , $T_{50R} \leq T_{\text{month}} < T_{\text{max}}$) and *Heat Exclusion* (purple, HE , $T_{\text{max}} \leq T_{\text{month}}$).

7. INTERACTIVE MAP

We provide the following link to the repository containing the data deposit and anonymized for an improved, interactive visualization of Figure 2a in the main text:

→ [https://macro-therm.github.io/pests_r_warming/]

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